



Onboarding Databases to DSF Hub Reference Guide

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Azure Data Sources

DSF Hub supports the following Azure data sources. Please select a link below to view the step-by-step instructions for each specific platform.

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- [Azure Cosmos DB Data Sources](#)
 - [Azure Cosmos DB for MongoDB Onboarding Steps](#)
 - [Azure Cosmos DB for NoSQL Onboarding Steps](#)
 - [Azure Cosmos DB for Table Onboarding Steps](#)
- [Azure Database for MariaDB Onboarding Steps](#)
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- [Azure Data Lake Storage Gen2 Onboarding Steps](#)
- [Azure Dedicated SQL Pool Onboarding Steps](#)
- [Azure SQL Managed Instance Onboarding Steps](#)
- [Azure SQL Server Onboarding Steps](#)

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Azure Blob Storage Onboarding Steps

This topic reviews the steps required to audit activity on Azure Blob Storage and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.
- You have permissions to create diagnostic settings for Azure Blob Storage to configure audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the database.
- You have a Container within a Storage account for DSF metadata storage.

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Specify the networks permitted to access the account. This can be done from the Advanced tab when creating the service, or by navigating to the Settings menu on your storage account. See the [Azure documentation](#) for more details.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

1. In the Azure portal, select Storage accounts and click on the name of the storage account to open the menu.
2. Select Diagnostic settings (classic) in the Monitoring (classic) section of the menu. For more details see the [Azure documentation](#).
 - a. Toggle the "Status" to "On".
 - b. Enable logging, metrics and retention policy for the data.
 - c. Click "Save" to finish.
3. Select Diagnostic settings in the Monitoring section of the menu.
 - a. Under "Name" click on "blob".
 - b. Click on "+ Add diagnostic setting" to add a new diagnostic.
 - c. Under "Logs" → "Categories" check the options: StorageRead, StorageWrite, StorageDelete.
 - d. Under "Destination details" check the "Stream to an event hub" and from the drop-down list select the corresponding values for: subscription, event hub namespace, event hub name, eventhub policy name (default value is RootManageSharedAccessKey).
 - e. Click "Save" to finish.

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Blob Storage Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
 2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
-

3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

- Asset template name for Azure Blob Storage: `AZURE_BLOB_STORAGE_template.xlsx`
- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

1. Gateway service log files:
 - a. Sonarsyslog service log file: `${JSONAR_LOGDIR}/gateway/syslog/storage_blob_eventhub.log`
 - b. Azure Eventhub service log file: `${JSONAR_LOGDIR}/eventhub/eventhub.log`

-
2. Run the following commands to verify the status of the various services:

```
systemctl status -l sonarrsyslog  
systemctl status -l sonareventhub
```

3. After the audit is turned on:
- On the Azure portal, select your Event Hub and go to the "Messages" dashboard.
 - Select a time frame period and check the count under the "Outgoing messages" tag.
 - If the count is greater than 0, in the selected time frame, this indicates that messages are being received by the Event Hub.

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

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Azure Cosmos DB Data Sources

DSF Hub supports the following Azure Cosmos DB APIs as a data source. Audit logs are generated by enabling the native Azure diagnostic logging feature. These logs are then directed to an Azure Event Hub, from which the Agentless Gateway ingests them. Note that each API may support a different level of auditing, depending on the operations and events tracked by the respective API.

Please select a link below to view the step-by-step instructions for each API.

- [Azure Cosmos DB for MongoDB Onboarding Steps](#)
- [Azure Cosmos DB for NoSQL Onboarding Steps](#)
- [Azure Cosmos DB for Table Onboarding Steps](#)

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Azure Cosmos DB for MongoDB Onboarding Steps

This topic reviews the steps required to audit activity on Azure Cosmos DB for MongoDB databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Audit Data Scope

The scope of audit data collection varies between data sources. Prior to enabling auditing, read through the auditing scope for this data source to identify any potential benefits or constraints. For more information on data collections, see [Main Collections in sonargd Database](#).

- Audit events from this data source are stored in the **instance**, **exception**, and (optionally) **slow_query** collections.
- Since the username is not included in audit logs, DSF Hub maps the `userId` value to the `DB User Name` field. If `userId` is not available, `DB User Name` defaults to "N/A".
- The following failed actions are recorded as successful documents:
 - Failed `ControlPlaneRequest` actions. The audit logs contain no error code or message, so these cannot be distinguished from successful requests.
 - Failed `db.getLastError()` calls. The audit logs record these as successful even when the CLI returns an error.
- The following information is missing from the audit logs:
 - `Analyzed Client IP` and `Client Host Name` are missing for `ControlPlaneRequests`.
 - `Source Program` is missing for actions from Azure CLI.
 - SQL string that caused the exception is not included in the audit logs.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps.

Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- You have permissions to create diagnostic settings in your Cosmos DB Account to configure audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the data source.
- You have a Container within a Storage Account for DSF metadata storage.
- You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Note:

- The different API offerings of Cosmos DB such as Cosmos DB for NoSQL, Cosmos DB for Table and Cosmos DB for MongoDB are all treated as distinct data sources. Each data source must send audit data to a unique Event Hub in any desired namespace.
- The audit data of multiple Cosmos DB instances may be sent to a single Event Hub provided they are of the same Server Type.

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

1. Sign into the Azure portal and navigate to your Azure Cosmos DB account.
2. Under the "Monitoring" section in the sidebar, select "Diagnostic settings".
3. Create a new diagnostic setting or edit an existing one.
 - a. To create a new diagnostic setting click on "+ Add diagnostic setting" and provide a name for the diagnostic setting.
 - b. To update an existing diagnostic setting, click on "Edit setting".
4. Under the "Logs" section to the left, select the following categories:
 - a. MongoRequests
 - b. ControlPlaneRequests
5. Under the "Destination details" section, click on "Stream to an event hub" checkbox.
6. From the "Event hub namespace" dropdown, choose your Event Hub namespace.
7. From the "Event hub name (optional)" dropdown, select the Event Hub created in Step 1 "Gathering Permissions & Information" in your chosen Event Hub namespace.

Note:

It is not recommended to choose the "(Create in selected namespace)" dropdown option which will create one Event Hub per selected log category in the chosen Event Hub namespace. If you choose to do so, when importing the assets in Step 3 "Collecting Audit Data", create one Event Hub asset with a unique Azure Storage Account Container per auto-generated Event Hub to audit all diagnostic log data.

Selecting an individual Event Hub to which to stream the diagnostic logs will allow the use of only one Azure Storage Account Container and the import of only one Event Hub asset in Step 3.

8. From the "Event hub policy name" dropdown, select "RootManageSharedAccessPolicy".
9. Select Save.

For more information, see [Monitor Azure Cosmos DB data using Azure Monitor Log Analytics diagnostic settings](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Cosmos DB Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the

dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click **Advanced**.

6. Under **Monitoring**, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Note:

While creating the Azure EventHub asset, select the "Cosmos Mongo" Audit Type.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under **Monitoring**, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

To import the Azure Cosmos DB for MongoDB asset, two assets are required: an Azure Cosmos DB for MongoDB asset and an Azure Event Hub asset.

- Asset template spreadsheet name for Azure Cosmos DB for MongoDB:
AZURE_COSMOS_MONGO_template.xlsx
- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template spreadsheet name for Azure account; choose ONE of the 3 templates listed below based on the type of authentication mechanism:

- **AZURE_MANAGED_IDENTITY_template.xlsx**
- **AZURE_AUTH_FILE_template.xlsx**
- **AZURE_CLIENT_SECRET_template.xlsx**

Discovery is currently only supported for Request unit (RU) database accounts.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

1. Gateway service log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/cosmosdb_mongo_eventhub.log
```

2. Run the following commands to verify the status of the gateway services and restart if necessary:

```
sudo systemctl status -l sonareventhub
sudo systemctl status -l sonareventhub.timer
sudo systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure Cosmos DB for NoSQL Onboarding Steps

This topic reviews the steps required to audit activity on Azure Cosmos DB NoSQL databases and send that information to DSF Hub for auditing, analyzing, detecting, and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

-
- An overview of the available scope for the data source, given its native audit data.
 - A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
 - Instructions on how to enable audit on the data source and collect data using DSF Hub.
 - Initial troubleshooting steps and technical support information.

Audit Data Scope

The scope of audit data collection varies between data sources. Prior to enabling auditing, read through the auditing scope for this data source to identify any potential benefits or constraints. For more information on data collections, see [Main Collections in sonargd Database](#).

- Audit events from this data source are stored in the **instance** and **exception** collections.
- Because the username is not included in the audit log, DSF Hub maps the `userId` value to the `DB User Name` field. If the `user ID` is not available, the `DB User Name` field will default to "N/A".
- For most actions, the audit log only shows what type of action was performed (such as Update, Insert, or Read) — not the actual data that was changed or accessed
- The `Objects and Verbs` and `Original SQL` fields are built by combining values from the raw log data, since they are not directly provided by the source.
- The following use cases are not audited:
 - Most user and role management actions (except when adding, changing, or removing access permissions).
 - Exporting data.
- The following information is missing from the audit logs:
 - `Analyzed Client IP` and `Client Host Name` are missing for `ControlPlaneRequests` and `QueryRuntimeStatistics` audit types.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- You have permissions to create diagnostic settings in your Cosmos DB Account to configure audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the data source.
- You have a Container within a Storage Account for DSF metadata storage.
- You have access to the Event Hub and storage account secret keys for pulling the audit logs.

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Note:

- The different API offerings of Cosmos DB such as Cosmos DB for NoSQL, Cosmos DB for Table and Cosmos DB for MongoDB are all treated as distinct data sources. Each data source should send audit data to a unique Event Hub in any desired namespace.

- The audit data of multiple Cosmos DB instances may be sent to a single Event Hub provided they are of the same Server Type.

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

1. Sign into the Azure portal and navigate to your Azure Cosmos DB account.
2. Under the "Monitoring" section in the sidebar, select "Diagnostic settings".
3. Create a new diagnostic setting or edit an existing one.
 - a. To create a new diagnostic setting click on "+ Add diagnostic setting" and provide a name for the diagnostic setting.
 - b. To update an existing diagnostic setting, click on "Edit setting".
4. Under the "Logs" section to the left, select the following categories:

Note:

QueryRuntimeStatistics audit plane provides the raw SQL query. By default, the query text and its parameters are obfuscated to avoid logging personal data (see [Monitor Azure Cosmos DB data using Azure Monitor Log Analytics diagnostic settings](#)) but full-text query logging can be enabled to show the full query in the audit log. For instructions on how to enable this, see [Enable full-text query for logging query text](#).

- a. DataPlaneRequests

- b. ControlPlaneRequests
- c. QueryRuntimeStatistics
5. Under the "Destination details" section, click on "Stream to an event hub" checkbox.
6. From the "Event hub namespace" dropdown, choose your Event Hub namespace.
7. From the "Event hub name (optional)" dropdown, select the Event Hub created in Step 1 "Gathering Permissions & Information" in your chosen Event Hub namespace.

Note:

It is not recommended to choose the "(Create in selected namespace)" dropdown option which will create one Event Hub per selected log category in the chosen Event Hub namespace. If you choose to do so, when importing the assets in Step 3 "Collecting Audit Data", create one Event Hub asset with a unique Azure Storage Account Container per auto-generated Event Hub to audit all diagnostic log data.

Selecting an individual Event Hub to which to stream the diagnostic logs will allow the use of only one Azure Storage Account Container and the import of only one Event Hub asset in Step 3.

8. From the "Event hub policy name" dropdown, select "RootManageSharedAccessPolicy".
9. Select Save.

For more information, see [Monitor Azure Cosmos DB data using Azure Monitor Log Analytics diagnostic settings](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Cosmos DB Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Note:

While creating the Azure EventHub asset, select the "Cosmos_SQL" Audit Type.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.

- Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
- Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

- Locate the desired cloud account asset in USC as described above.
- Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
- Click **Start Discovery**. Discovery will start running in the background.
- After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

- From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
- On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
- Use the Asset Specification documentation as a guide for filling in the necessary field values.
- On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
- To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
- Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
- Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
- Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

To import an Azure Cosmos DB for NoSQL asset, two assets are required: an Azure Cosmos DB for NoSQL asset and an Azure Event Hub asset.

- Asset template spreadsheet name for Azure Cosmos DB for NoSQL:
 - DSF versions 4.18 and lower: **AZURE_COSMOS_SQL_template.xlsx**
 - DSF versions 4.19 and greater: **AZURE_COSMOS_NOSQL_template.xlsx**
- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared

access policies. The default name used by Azure is "RootManageSharedAccessKey".

- **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
- **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template spreadsheet name for Azure account; choose ONE of the 3 templates listed below based on the type of authentication mechanism:

- [AZURE_MANAGED_IDENTITY_template.xlsx](#)
- [AZURE_AUTH_FILE_template.xlsx](#)
- [AZURE_CLIENT_SECRET_template.xlsx](#)

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

1. Gateway service log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/cosmosdb_sql_eventhub.log
```

2. Run the following commands to verify the status of the gateway services and restart if necessary:

```
sudo systemctl status -l sonareventhub
sudo systemctl status -l sonareventhub.timer
sudo systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure Cosmos DB for Table Onboarding Steps

This topic reviews the steps required to audit activity on Azure Cosmos DB for Table databases and send that information to DSF Hub for auditing, analyzing, detecting, and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Audit Data Scope

The scope of audit data collection varies between data sources. Prior to enabling auditing, read through the auditing scope for this data source to identify any potential benefits or constraints. For more information on data collections, see [Main Collections in sonargd Database](#).

- Audit events from this data source are stored in the **instance** and **exception** collections.
- For most actions, the audit log may generate multiple entries — this is expected behavior, as the system also logs internal background operations that run alongside your action.
- The `Objects` and `Verbs` and `Original SQL` fields are built by combining values from the raw log data, since they are not directly provided by the source.
- Because the username is not included in the audit log, DSF Hub maps the `userId` value to the `DB User Name` field. If the `user ID` is not available, the `DB User Name` field will default to "N/A".
- The following use cases are not audited:
 - A failed attempt to delete a table (whether the table exists or not) through the GUI.
 - Checking whether a table exists, listing tables, or displaying table details through the Azure CLI.
- The following information is missing from the audit logs:
 - `Analyzed Client IP` and `Client Host Name` are missing for `ControlPlaneRequests` category.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- You have permissions to create diagnostic settings in your Cosmos DB Account to configure audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the data source.
- You have a Container within a Storage Account for DSF metadata storage.
- You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Note:

- The different API offerings of Cosmos DB such as Cosmos DB for NoSQL, Cosmos DB for Table and Cosmos DB for MongoDB are all treated as distinct data sources. Each data source should send audit data to a unique Event Hub in any desired namespace.
- The audit data of multiple Cosmos DB instances may be sent to a single Event Hub provided they are of the same Server Type.

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

1. Sign into the Azure portal and navigate to your Azure Cosmos DB account.
2. Under the "Monitoring" section in the sidebar, select "Diagnostic settings".
3. Create a new diagnostic setting or edit an existing one.
 - a. To create a new diagnostic setting click on "+ Add diagnostic setting" and provide a name for the diagnostic setting.
 - b. To update an existing diagnostic setting, click on "Edit setting".
4. Under the "Logs" section to the left, select the **DataPlaneRequests** log category.

Note:

Optionally, ControlPlaneRequests and TableApiRequests may also be selected.

5. Under the "Destination details" section, click on "Stream to an event hub" checkbox.
6. From the "Event hub namespace" dropdown, choose your Event Hub namespace.
7. From the "Event hub name (optional)" dropdown, select the Event Hub created in Step 1 (Gathering Permissions & Information) in your chosen Event Hub namespace.

Note:

It is not recommended to choose the "(Create in selected namespace)" dropdown option when more than one log category is selected because it will create one Event Hub per selected log category in the chosen Event Hub namespace. If this option is selected, in Step 3 "Collecting Audit Data", it is required to create an Event Hub asset with a unique Azure Storage Account Container per auto-generated Event Hub to audit all diagnostic log data.

Selecting an individual Event Hub in which to stream all the diagnostic logs allows the use of only one Azure Storage Account Container and the import of only one Event Hub asset in Step 3.

8. From the "Event hub policy name" dropdown, select "RootManageSharedAccessPolicy".
9. Select Save.

For more information, see [Monitor Azure Cosmos DB data using Azure Monitor Log Analytics diagnostic settings](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Cosmos DB Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Note:

While creating the Azure EventHub asset, select the "Cosmos Table" Audit Type.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the

cloud account.

- For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
 7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

To import the Azure Cosmos DB for Table asset, two assets are required: an Azure Cosmos DB for Table asset and an Azure Event Hub asset.

- Asset template spreadsheet name for Azure Cosmos DB for Table:
AZURE_COSMOS_TABLE_template.xlsx

- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template spreadsheet name for Azure account; choose ONE of the 3 templates listed below based on the type of authentication mechanism:

- **AZURE_MANAGED_IDENTITY_template.xlsx**
- **AZURE_AUTH_FILE_template.xlsx**
- **AZURE_CLIENT_SECRET_template.xlsx**

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

1. Gateway service log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/cosmosdb_table_eventhub.log
```

2. Run the following commands to verify the status of the gateway services and restart if necessary:

```
sudo systemctl status -l sonareventhub
sudo systemctl status -l sonareventhub.timer
sudo systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.

-
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure Database for MariaDB Onboarding Steps

This topic reviews the steps required to audit activity on Azure Database for MariaDB and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- If using manual import: You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.
- If using Discovery: A client secret, auth_file or managed identity is needed for DSF to access Azure resources.
- You have permissions to create diagnostic settings in your Azure Database for MariaDB to configure audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the database.
- You have a Container within a Storage Account for DSF metadata storage.

Authentication Mechanisms for Discovery with Azure Cloud Account Asset

DSF supports the following authentication mechanisms for accessing Azure resources:

- Client secret: for detailed instructions on creating a client secret, please see [Register a Microsoft Entra app and create a service principal](#) and [Option 3: Create a new client secret](#).
- Managed Identity: for detailed instructions on managed identities, please see [Configure managed identities on Azure virtual machines \(VMs\)](#).
- Auth file: a JSON file that contains the client secret credentials (see an example in the [Azure Asset Specifications](#)). Place this file in the `$JSONAR_LOCALDIR/credentials` directory on the Agentless Gateway, and ensure that its absolute path is specified in the “key_file” field of the Azure cloud account connection document.

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

```
Microsoft.Authorization/permissions/read
Microsoft.EventHub/namespaces/authorizationrules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/write
Microsoft.EventHub/namespaces/eventhubs/read
Microsoft.EventHub/namespaces/eventhubs/write
Microsoft.EventHub/namespaces/read
Microsoft.EventHub/namespaces/write
Microsoft.Storage/storageAccounts/listkeys/action
Microsoft.Storage/storageAccounts/read
Microsoft.Storage/storageAccounts/write
Microsoft.Insights/DiagnosticSettings/Read
Microsoft.Insights/DiagnosticSettings/Write
Microsoft.DBforMariaDB/servers/read
```

If you intend to use DSF's Audit Policy Management feature, then the following additional privileges should also be granted:

```
Microsoft.DBForMariaDB/servers/configurations/read
Microsoft.DBForMariaDB/servers/configurations/write
Microsoft.Insights/diagnosticSettings/delete
```

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is

not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.

- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers

and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.
- Additionally, ensure the following outbound ports are also open on the Agentless Gateway:
 - 3306: to connect to the database for turning on audit policies

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

This documentation provides instruction to enable audit using both Azure Portal and CLI. Please enable audit using any either method.

- **Enable Auditing using Azure Portal**
(or)
- **Enable Auditing using Azure CLI**

Enable Auditing using Azure Portal

Configure audit logging by following these steps:

1. Sign in to the Azure portal.
2. Select your Azure Database for MariaDB server.
3. Under the Settings section in the sidebar, select Server parameters.
4. Update the `audit_log_enabled` parameter to ON.
5. Select the event types to be logged by updating the `audit_log_events` parameter. To audit all actions on the database select "CONNECTION" and "GENERAL" events from the drop-down.
6. It is recommended to update the `audit_log_exclude_users` parameter to exclude 'azure_superuser'. This is a system user created by Microsoft to manage the server to conduct monitoring, backups, and other regular maintenance that may be considered "noise".
7. Once you have changed the parameters, select Save.

For more information on configuring audit logging using the Azure Portal, see [Configure Audit Logging](#).

Set up diagnostic logs by following these steps:

1. Under the Monitoring section in the sidebar, select Diagnostic settings.
2. Select on "+ Add diagnostic setting".
3. Provide a diagnostic setting name.
4. Click on "Stream to an event hub" check box and choose your event hub namespace and event hub from the drop-down.
5. Select "MariaDB Audit Logs" as the log type.
6. Select Save.

For more information on configuring Diagnostic settings using the Azure Portal, see [Set up Diagnostic logs](#).

Enable Auditing using Azure CLI

Configure audit logging using the following steps. Specify your own Resource Group and Server name, all other values should stay the same.

1. Turn on audit logs by setting the `audit_logs_enabled` parameter to "ON".

```
az mariadb server configuration set --name audit_log_enabled --resource-group myresourcegr
```

2. Select the event types to be logged by updating the `audit_log_events` parameter.

```
az mariadb server configuration set --name audit_log_events --resource-group myresourcegr
```

3. It is recommended to update the `audit_log_exclude_users` parameter to exclude 'azure_superuser'. This is a system user created by Microsoft to manage the server to conduct monitoring, backups, and other regular maintenance that may be considered "noise".

```
az mariadb server configuration set --name audit_log_exclude_users --resource-group myres
```

For more information refer to Azure product documentation [Configure and access Azure Database for MariaDB audit logs in the Azure CLI](#)

Set up diagnostic logs

Use the following command to setup a diagnostic setting to stream audit logs to an eventhub. In the command below, specify the diagnosticname, my-subscription-id, myresourcegroup, my-mariadb-server, myeventhubnamespace and my-eventhub values. All other values should stay the same.

```
az monitor diagnostic-settings create \
--name diagnosticname \
--resource /subscriptions/my-subscription-id/resourceGroups/myresourcegroup/providers/Micros
--logs '[{"category": "MariaDB Audit Logs", "enabled": true}]' \
--event-hub /subscriptions/my-subscription-id/resourceGroups/myresourcegroup/providers/Micro
--event-hub-rule /subscriptions/my-subscription-id/resourceGroups/myresourcegroup/providers/
```

For more information refer to Azure product documentation [Azure CLI diagnostic-settings](#).

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Asset Specifications](#)

[Azure MariaDB Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.

4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data](#)

Sources in DSF Hub Cloud Accounts using USC documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

- Asset template name for Azure Database for MariaDB: **AZURE_MARIADB_template.xlsx**
- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template name for Azure account; choose ONE of the 3 templates listed below based on the type of authentication mechanism:

- **AZURE_MANAGED_IDENTITY_template.xlsx**
- **AZURE_AUTH_FILE_template.xlsx**
- **AZURE_CLIENT_SECRET_template.xlsx**

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

✓STEP 4: Managing Audit Policies

Audit Policies are created to specify which actions will be collected from the database. This can be done in two ways:

- **Using the Audit Policy Management Feature**
(or)
- **Creating an Audit Policy manually**

Using the Audit Policy Management Feature

DSF provides automated solutions for managing and monitoring audit policies. Audit policies can be managed from the Asset Dashboard (OR) USC.

Managing Audit Policies with Asset Dashboard

To apply an audit policy via Asset Dashboard:

1. From the **DSF Portal**, expand the left-side navigation menu by clicking **>>**. Click **Data Activity Monitoring > Search & Analyze > Dashboard**. Alternatively, you can use the search bar to locate **Dashboard**.
2. On the **Dashboards** page, select the **Asset Details** dashboard.
3. In the **Asset Details** dashboard in DSF DataVis, go to the **Assets** panel and locate the asset to which you want to apply the audit policy.
4. Click the three vertical dots at the end of the row for the asset, in the **Actions** column.
5. In the menu that opens, select **Management > Audit Actions > Turn on Audit Policy**. The **Turn On Audit Policy** window opens.
6. In the **Turn On Audit Policy** window, under **Audit Policy**, select an audit policy by type or by name.
7. Click **Submit**. The audit policy is turned on, that is, applied to the asset.
8. If you want, click **Refresh** to view the asset's audit policy details (e.g. policy_template_name) in the relevant row in the **Assets** panel.

For detailed instructions and supported database versions, see [Managing Audit Policies](#) in the user guide.

Managing Audit Policies with Unified Settings Console

To create and apply a new DSF Hub audit policy via USC:

1. From the main page of **DSF Portal**, under **Apps**, click **Unified Settings Console**.
2. In the **Appliances** pane, select **DSF Hub**.
3. Click the **Audit Policies** tab. The audit policies of the DSF Hub appliance are displayed.
4. Click **Add**.
5. In the **Configuration** tab, in the **Policy Name** field, type a name for the policy.
6. In the **Configuration** tab, select a policy type. Options available in this menu include **Native audit policy**, **Gateway audit policy** and **Combined audit policy**.
7. Select match criteria. Available options depend on audit type selected: For more information about these criteria, see [Supported Criteria for Audit Policies](#).

8. If you selected the **Exclude List** type, add at least one of the following sets:
 - DB User Name
 - Analyzed Client IP
 - Source Program
 You can add multiple sets of **Exclude List** criteria via the **+ Add** button.
9. If you selected the **Action Groups** policy type, add one or more action groups. For more information, see [Viewing Lists of Actions for Action Groups](#).
10. Click the **Applied to** tab and select the data source assets to which the policy will be applied.
11. Turn **Audit collection** on or off. This option provides a convenient method to enable audit collection directly from the audit policy.
12. Click **Save**. DSF Hub begins updating the audit policy; this process may take a few minutes to complete.

For detailed instructions, see [Managing Audit Policies via USC](#).

Note:

- Creating Audit policies via USC is not supported for this data source.
- DSF Audit Management feature audits all actions by enabling **"GENERAL"** and **"CONNECTION"** events on Azure Database for MariaDB

Creating an Audit Policy Manually

Alternatively, it is possible to manually create and manage audit policies on the data source itself.

1. Log in to your Azure Portal.
2. Select your Azure Database for MariaDB server.
3. Under the **Settings** section in the sidebar, select **Server parameters**.
4. Select the **event types** to be logged by updating the **audit_log_events** parameter. To audit all actions on the database select **"CONNECTION"** and **"GENERAL"** events from the drop-down.
5. Once all the parameters are updated, select **Save**.

For more information on configuring audit policy using the Azure Portal, see [Audit policy configuration](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

1. Gateway service log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/mariadb_eventhub.log
```

2. Run the following commands to verify the status of the gateway services and restart if necessary:

```
sudo systemctl status -l sonareventhub
sudo systemctl status -l sonareventhub.timer
sudo systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure Database for MySQL Single Server Onboarding Steps

This topic reviews the steps required to audit activity on Azure Database for MySQL databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Note: Azure Database for MySQL includes two deployment options: **Single Server** and **Flexible Server**. This page provides onboarding instructions for **Single Server** only. For **Flexible Server**, see [LINK](#).

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The

following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- If using manual import: You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.
- If using Discovery: A client secret, auth_file or managed identity is needed for DSF to access Azure resources.
- You have permissions to create diagnostic settings in your Azure Database for MySQL to configure audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the database.
- You have a Container within a Storage Account for DSF metadata storage.

Authentication Mechanisms for Discovery with Azure Cloud Account Asset

DSF supports the following authentication mechanisms for accessing Azure resources:

- Client secret: for detailed instructions on creating a client secret, please see [Register a Microsoft Entra app and create a service principal](#) and [Option 3: Create a new client secret](#).
- Managed Identity: for detailed instructions on managed identities, please see [Configure managed identities on Azure virtual machines \(VMs\)](#).
- Auth file: a JSON file that contains the client secret credentials (see an example in the [Azure Asset Specifications](#)). Place this file in the `$JSONAR_LOCALDIR/credentials` directory on the Agentless Gateway, and ensure that its absolute path is specified in the "key_file" field of the Azure cloud account connection

document.

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

```
Microsoft.Authorization/permissions/read
Microsoft.EventHub/namespaces/authorizationrules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/write
Microsoft.EventHub/namespaces/eventhubs/read
Microsoft.EventHub/namespaces/eventhubs/write
Microsoft.EventHub/namespaces/read
Microsoft.EventHub/namespaces/write
Microsoft.Storage/storageAccounts/listkeys/action
Microsoft.Storage/storageAccounts/read
Microsoft.Storage/storageAccounts/write
Microsoft.Insights/DiagnosticSettings/Read
Microsoft.Insights/DiagnosticSettings/Write
Microsoft.DBforMySQL/servers/read
```

If you intend to use DSF's Audit Policy Management feature, then the following additional privileges should also be granted:

```
Microsoft.DBforMySQL/servers/configurations/read
Microsoft.DBforMySQL/servers/configurations/write
Microsoft.Insights/diagnosticSettings/delete
```

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

-
- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
 - When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
 - Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
 - The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.
 - Additionally, ensure the following outbound ports are also open on the Agentless Gateway:
 - 3306: to connect to the database for turning on audit policies

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

This documentation provides instructions for enabling audit via either **DSF**, or via the **Azure Portal** or **CLI**. Please choose only one of the three methods.

Enable Auditing via DSF

Audit can be enabled via DSF by using the DSF Audit Management feature. No further action is required here, please skip to Step 3 "Collecting Audit Data".

Enable Auditing via Azure

Enable Auditing using Azure Portal

To configure audit logging follow these steps:

1. Sign in to the Azure portal.
2. Select your Azure Database for MySQL server.
3. Under the Settings section in the sidebar, select Server parameters.
4. Update the audit_log_enabled parameter to ON.
5. Select the event types to be logged by updating the audit_log_events parameter. To audit all actions on the database select "CONNECTION", "GENERAL" and "TABLE ACCESS" events from the drop-down.

6. It is recommended to update the `audit_log_exclude_users` parameter to exclude 'azure_superuser'. This is a system user that manages the server by conducting monitoring, backups, and other regular maintenance that may be considered "noise".
7. Once you have changed the parameters, select Save.

For more information on configuring audit logging using the Azure Portal, see [Configure Audit Logging](#).

Set up diagnostic logs

1. Under the Monitoring section in the sidebar, select Diagnostic settings.
2. Select on "+ Add diagnostic setting".
3. Provide a diagnostic setting name.
4. Click on "Stream to an event hub" check box and choose your event hub namespace and event hub from the drop-down.
5. Select "MySQL Audit Logs" as the log type.
6. Select Save.

For more information on configuring Diagnostic settings using the Azure Portal, see [Set up Diagnostic logs](#).

Enable Auditing using Azure CLI

Configure audit logging using the following steps. Specify the resource-group and mysql-server. All other values should remain the same.

1. Turn on audit logs by setting the `audit_logs_enabled` parameter to "ON".

```
az mysql server configuration set --name audit_log_enabled --resource-group <resource-group>
```

2. Select the event types to be logged by updating the `audit_log_events` parameter.

```
az mysql server configuration set --name audit_log_events --resource-group <resource-group>
```

3. It is recommended to update the `audit_log_exclude_users` parameter to exclude 'azure_superuser'. This is a system user created by Microsoft to manage the server to conduct monitoring, backups, and other regular maintenance that may be considered "noise".

```
az mysql server configuration set --name audit_log_exclude_users --resource-group <resource-group>
```

For more information refer to Azure product documentation [Configure and access Azure Database for MySQL audit logs via the Azure CLI](#).

Set up diagnostic logs

Use the following command to setup a diagnostic setting to stream audit logs to an Event Hub. In the command below, specify the diagnostic-name, subscription-id, resource-group, mysql-server, eventhub-namespace and eventhub values. All other values should remain the same.

```
az monitor diagnostic-settings create \
--name <diagnostic-name> \
--resource /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/MicrosoftMonitoring/providers/MicrosoftMonitoring/diagnosticSettings/<diagnostic-name> \
--logs '[{"category": "MySQL Audit Logs", "enabled": true}]' \
--event-hub /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/MicrosoftMonitoring/providers/MicrosoftMonitoring/eventHubs/<event-hub-name> \
--event-hub-rule /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/MicrosoftMonitoring/providers/MicrosoftMonitoring/eventHubs/<event-hub-name>/rules/<event-hub-rule-name>
```

For more information refer to Azure product documentation [Azure CLI diagnostic-settings](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Asset Specifications](#)

[Azure MySQL Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your

asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
 2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
 3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
-

4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

- Asset template name for Azure Database for MySQL: **AZURE_MYSQL_template.xlsx**
- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.

6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template name for Azure account; choose ONE of the 3 templates below based on the type of authentication mechanism:

- **AZURE_MANAGED_IDENTITY_template.xlsx**
- **AZURE_AUTH_FILE_template.xlsx**
- **AZURE_CLIENT_SECRET_template.xlsx**

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

✓STEP 4: Managing Audit Policies

Audit Policies are created to specify which actions will be collected from the database. This can be done in two ways:

- **Using the Audit Policy Management Feature**
(or)
- **Creating an Audit Policy manually**

Using the Audit Policy Management Feature

DSF provides automated solutions for managing and monitoring audit policies. Audit policies can be managed from the Asset Dashboard (OR) USC.

Managing Audit Policies with Asset Dashboard

To apply an audit policy via Asset Dashboard:

1. From the **DSF Portal**, expand the left-side navigation menu by clicking >>. Click **Data Activity Monitoring > Search & Analyze > Dashboard**. Alternatively, you can use the search bar to locate **Dashboard**.

2. On the **Dashboards** page, select the **Asset Details** dashboard.
3. In the **Asset Details** dashboard in DSF DataVis, go to the **Assets** panel and locate the asset to which you want to apply the audit policy.
4. Click the three vertical dots at the end of the row for the asset, in the **Actions** column.
5. In the menu that opens, select **Management > Audit Actions > Turn on Audit Policy**. The **Turn On Audit Policy** window opens.
6. In the **Turn On Audit Policy** window, under **Audit Policy**, select an audit policy by type or by name.
7. Click **Submit**. The audit policy is turned on, that is, applied to the asset.
8. If you want, click **Refresh** to view the asset's audit policy details (e.g. policy_template_name) in the relevant row in the **Assets** panel.

For detailed instructions and supported database versions, see [Managing Audit Policies](#) in the user guide.

Managing Audit Policies with Unified Settings Console

To create and apply a new DSF Hub audit policy via USC:

1. From the main page of **DSF Portal**, under **Apps**, click **Unified Settings Console**.
2. In the **Appliances** pane, select **DSF Hub**.
3. Click the **Audit Policies** tab. The audit policies of the DSF Hub appliance are displayed.
4. Click **Add**.
5. In the **Configuration** tab, in the **Policy Name** field, type a name for the policy.
6. In the **Configuration** tab, select a policy type. Options available in this menu include **Native audit policy**, **Gateway audit policy** and **Combined audit policy**.
7. Select match criteria. Available options depend on audit type selected: For more information about these criteria, see [Supported Criteria for Audit Policies](#).
8. If you selected the **Exclude List** type, add at least one of the following sets:
 - DB User Name
 - Analyzed Client IP
 - Source Program

You can add multiple sets of **Exclude List** criteria via the **+ Add** button.
9. If you selected the **Action Groups** policy type, add one or more action groups. For more information, see [Viewing Lists of Actions for Action Groups](#).

10. Click the **Applied to** tab and select the data source assets to which the policy will be applied.
11. Turn **Audit collection** on or off. This option provides a convenient method to enable audit collection directly from the audit policy.
12. Click **Save**. DSF Hub begins updating the audit policy; this process may take a few minutes to complete.

For detailed instructions, see [Managing Audit Policies via USC](#).

Note:

- Creating Audit policies via USC is not supported for this data source.
- DSF Audit Management feature audits all actions by enabling **"GENERAL"**, **"CONNECTION"** and **"TABLE ACCESS"** events on Azure Database for MySQL.

Creating an Audit Policy Manually

Alternatively, it is possible to manually create and manage audit policies on the data source itself.

If the auditing was enabled manually via the Azure Portal, as described in Step 2, then the `audit_log_events` parameter is already set to audit all actions: "CONNECTION", "GENERAL" and "TABLE ACCESS". No other action is required.

For more information on configuring audit policy using the Azure Portal, see [Audit policy configuration](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

1. Gateway service log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/mysql_eventhub.log
```

2. Run the following commands to verify the status of the gateway services and restart if necessary:

```
sudo systemctl status -l sonareventhub
sudo systemctl status -l sonareventhub.timer
sudo systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure Database for MySQL Flexible Server Onboarding Steps

This topic reviews the steps required to audit activity on Azure Database for MySQL Flexible Server databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✔STEP 1: Gathering Permissions & Information

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- You have permissions for modifying server parameters in your Azure Database for MySQL Flexible Server to configure audit.

If importing the assets to DSF manually:

- You have permissions for creating diagnostic settings in your Azure Database for MySQL Flexible Server to stream audit logs to an Event Hub.
- You have an Event Hub within an Event Hub Namespace for receiving and storing the audit logs from the database.
- You have a Container within a Storage account for DSF metadata storage.
- You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.

Authentication Mechanisms for Discovery with Azure Cloud Account Asset

DSF supports the following authentication mechanisms for accessing Azure resources:

- Client secret: for detailed instructions on creating a client secret, please see [Register a Microsoft Entra app and create a service principal](#) and [Option 3: Create a new client secret](#).
- Managed Identity: for detailed instructions on managed identities, please see [Configure managed identities on Azure virtual machines \(VMs\)](#).
- Auth file: a JSON file that contains the client secret credentials (see an example in the [Azure Asset Specifications](#)). Place this file in the \$JSONAR_LOCALDIR/credentials directory on the Agentless Gateway, and ensure that its absolute path is specified in the “key_file” field of the Azure cloud account connection document.

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

```
Microsoft.Authorization/permissions/read
Microsoft.EventHub/namespaces/authorizationrules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/write
Microsoft.EventHub/namespaces/eventhubs/read
Microsoft.EventHub/namespaces/eventhubs/write
Microsoft.EventHub/namespaces/read
Microsoft.EventHub/namespaces/write
Microsoft.Storage/storageAccounts/listkeys/action
Microsoft.Storage/storageAccounts/read
Microsoft.Storage/storageAccounts/write
Microsoft.Insights/DiagnosticSettings/Read
Microsoft.Insights/DiagnosticSettings/Write
Microsoft.DBforMySQL/flexibleServers/read
```

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:

When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.

- For DSF version 4.12:

Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: `<name-of-eventhub-namespace>.servicebus.windows.net`
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

This section provides instructions for enabling audit using either the **Azure Portal** or **CLI**. Please choose only one of the two methods.

Enable Auditing using Azure Portal

Enable Audit Logging on the MySQL Flexible Server

To configure audit logging follow these steps:

1. Sign in to the Azure portal.
2. Select your Azure Database for MySQL flexible server.
3. Under the Settings section in the sidebar, select Server parameters.
4. Update the **audit_log_enabled** parameter to **ON**.
5. Select the event types to be logged by updating the **audit_log_events** parameter. To audit all actions on the database, select "**CONNECTION_V2**", "**GENERAL**" and "**TABLE ACCESS**" events from the dropdown.

Note:

You must select **CONNECTION_V2** to generate the required session activity data. Choosing **CONNECTION** will not work, as the generated data will be missing timestamp values.

6. In the Server Parameters page, select the "**ALL**" tab.
7. Search for the parameter **audit_log_exclude_users** and set the value as "**azure_superuser**". This is a system user that runs internal actions and generates audit data that may be considered as noise. For more information on the `azure_superuser` user, see [About azure_superuser](#).
8. Once you have changed the parameters, select Save.

For more information on configuring audit logging using the Azure Portal, see [Configure Audit Logging](#).

Enable Slow Query Monitoring on the MySQL Flexible Server (Optional)

Note:

- Turning on Slow Query Monitoring will not interfere with standard audit collection.
- This step is **optional** and required only if you choose to audit slow query logs.
- Queries that exceed the minimum duration threshold will be mapped to a separate collection called `sonargd.slow_query`.

The slow query log consists of SQL statements that take more than a set threshold in seconds to execute. To enable slow query monitoring and set said threshold, follow these steps:

1. From the Azure portal, Navigate to your Azure Database for MySQL flexible server.
2. Under the Settings section in the sidebar, select Server parameters.
3. Set the **slow_query_log** parameter to **ON** to enable slow query monitoring.
4. Set the **long_query_time** parameter to your desired value in seconds. This is the duration threshold for the

query to be considered as a slow query.

5. Click **Save** to save your changes.

For more information on configuring slow query monitoring, see [Configure slow query logging](#).

Set up Diagnostic Settings to Stream Audit Data to an Event Hub

To create a diagnostic setting that would stream audit data to an Event Hub, follow these steps:

1. Under the Monitoring section in the sidebar, select Diagnostic settings.
2. Click on "+ Add diagnostic setting".
3. Provide a diagnostic setting name.
4. Under the "Destination details" section, click on the "Stream to an event hub" checkbox.
5. From the "Event hub namespace" dropdown, choose your Event Hub namespace.
6. From the "Event hub name (optional)" dropdown, select the Event Hub created in Step 1 "Gathering Permissions & Information" in your chosen Event Hub namespace.

Note:

It is not recommended to choose the "(Create in selected namespace)" dropdown option when slow query monitoring is enabled because it will create one Event Hub per selected log category in the chosen Event Hub namespace. If this option is selected, in Step 3 "Collecting Audit Data", it is required to create an Event Hub asset with a unique Azure Storage Account Container per auto-generated Event Hub to audit all diagnostic log data.

Selecting an individual Event Hub in which to stream all the diagnostic logs allows the use of only one Azure Storage Account Container and the import of only one Event Hub asset in Step 3.

7. From the "Event hub policy name" dropdown, select "RootManageSharedAccessPolicy".
8. Under the "Logs" section to the left, select **"MySQL Audit Logs"** log category.
 - a. If monitoring slow queries, select **"MySQL Slow Logs"** in addition to the above log category.
9. Select Save.

For more information on configuring Diagnostic settings using the Azure Portal, see [Tutorial: Configure audit logs by using Azure Database for MySQL - Flexible Server](#).

Enable Auditing using Azure CLI

Enable Audit Logging on the MySQL Flexible Server

To configure audit logging, follow the steps below. Be sure to replace '<resource-group>' and '<mysql-flexible-server>' with the names of your specific resource group and MySQL flexible server, respectively.

1. Turn on audit logs by setting the **audit_logs_enabled** parameter to **"ON"**.

```
az mysql flexible-server parameter set --resource-group <resource-group> --server-name <mysql-flexible-server>
```

2. Select the event types to be logged by updating the **audit_log_events** parameter.

```
az mysql flexible-server parameter set --resource-group <resource-group> --server-name <mysql-flexible-server> --audit-log-events <event-types>
```

3. It is recommended to update the **audit_log_exclude_users** parameter to exclude 'azure_superuser'. This is a system user that runs internal actions and generates audit data that may be considered as noise.

```
az mysql flexible-server parameter set --resource-group <resource-group> --server-name <mysql-flexible-server> --audit-log-exclude-users <users>
```

For more information on the `azure_superuser` user, see [About azure_superuser](#).

For more information, see [Configure audit logs on Azure Database for MySQL - Flexible Server using Azure CLI](#).

Enable Slow Query Monitoring on the MySQL Flexible Server (Optional)

To configure slow query monitoring, follow the steps below. Ensure to replace '<resource-group>' and '<mysql-flexible-server>' with the names of your specific resource group and MySQL flexible server, respectively.

1. Enable slow query monitoring by setting the **slow_query_log** server parameter to **ON**.

```
az mysql flexible-server parameter set --resource-group <resource-group> --server-name <mysql-flexible-server> --slow-query-log ON
```

2. Set the duration threshold for a query to be considered slow by configuring the `long_query_time` parameter. Replace <values> with your desired threshold in seconds.

```
az mysql flexible-server parameter set --resource-group <resource-group> --server-name <mysql-flexible-server> --long-query-time <values>
```

For more information on configuring slow query monitoring, see [Configure slow query logging](#).

Set up Diagnostic Settings to Stream Audit Data to an Event Hub

Use the following command to setup a diagnostic setting to stream audit logs to an Event Hub. In the command below, specify the diagnostic-name, subscription-id, resource-group, mysql-server, eventhub-namespace and eventhub values. All other values should remain the same.

```
az monitor diagnostic-settings create \
  --name <diagnostic-name> \
  --resource /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.DBForMySQL/flexibleServers/<mysql-server> \
  --logs '[{"category": "MySQL Audit Logs", "enabled": true}]' \
  --event-hub /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.EventHub/namespaces/<eventhub-namespace>/eventhubs/<eventhub> \
  --event-hub-rule /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.EventHub/namespaces/<eventhub-namespace>/eventhubs/<eventhub>/rules/<eventhub-rule>
```

For more information, see [Azure CLI diagnostic-settings](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Asset Specifications](#)

[Azure MySQL Flexible Server Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

To import the Azure Database for MySQL Flexible Server asset, two assets are required: an Azure Database for MySQL Flexible Server asset and an Azure Event Hub asset.

- Asset template name: **AZURE_MYSQL_FLEXIBLE_SERVER_template.xlsx**

In the spreadsheet template, you will need to provide the following details for the Event Hub asset:

- **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
- **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
- **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

- Gateway log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/mysql_flexible_eventhub.log
```

- Run the following commands to verify the status of the gateway services and restart if necessary:

```
sudo systemctl status -l sonareventhub
sudo systemctl status -l sonareventhub.timer
sudo systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure Database for PostgreSQL Single Server Onboarding Steps

This topic reviews the steps required to audit activity on Azure Database for PostgreSQL databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Note: Azure Database for PostgreSQL includes two deployment options: **Single Server** and **Flexible Server**. This page provides onboarding instructions for **Single Server** only. For **Flexible Server**, see [Azure Database for PostgreSQL Flexible Server Onboarding Steps](#).

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- If using manual import: You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.
- If using Discovery: A client secret, auth_file or managed identity is needed for DSF to access Azure resources.
- You have permissions to create diagnostic settings in your Azure Database for PostgreSQL to configure audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the database.
- You have a Container within a Storage Account for DSF metadata storage.

Authentication Mechanisms for Discovery with Azure Cloud Account Asset

DSF supports the following authentication mechanisms for accessing Azure resources:

- Client secret: for detailed instructions on creating a client secret, please see [Register a Microsoft Entra app and create a service principal](#) and [Option 3: Create a new client secret](#).
- Managed Identity: for detailed instructions on managed identities, please see [Configure managed identities on Azure virtual machines \(VMs\)](#).
- Auth file: a JSON file that contains the client secret credentials (see an example in the [Azure Asset Specifications](#)). Place this file in the \$JSONAR_LOCALDIR/credentials directory on the Agentless Gateway, and ensure that its absolute path is specified in the “key_file” field of the Azure cloud account connection document.

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

```
Microsoft.Authorization/permissions/read
Microsoft.EventHub/namespaces/authorizationrules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/write
Microsoft.EventHub/namespaces/eventhubs/read
Microsoft.EventHub/namespaces/eventhubs/write
Microsoft.EventHub/namespaces/read
Microsoft.EventHub/namespaces/write
Microsoft.Storage/storageAccounts/listkeys/action
Microsoft.Storage/storageAccounts/read
Microsoft.Storage/storageAccounts/write
Microsoft.Insights/DiagnosticSettings/Read
Microsoft.Insights/DiagnosticSettings/Write
Microsoft.DBforPostgreSQL/servers/read
```

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless

Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

Database Prerequisites

Below are the requirements for granting database permissions for audit log retrieval. Please ensure these are completed by a Database Administrator.

- You have a General Purpose or Memory Optimized Azure PostgreSQL server. Only these types of Azure Database for PostgreSQL servers support enabling of the pgAudit extension, which is a prerequisite to collect audit.
- You can connect to your Azure Database for PostgreSQL server, using a standard client (e.g. psql), with an admin user that also has "pg_read_all_settings" permissions, for creating and verifying the creation of the pgaudit extension.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

This documentation provides instructions to enable audit using either Azure Portal and CLI. Please select either method and refer to the [Audit logging in Azure Database for PostgreSQL - Single Server](#) documentation for more details.

Via Azure Portal

1. Select your Azure Database for PostgreSQL server.
2. On the left, under Settings, select Server parameters.
3. Search for shared_preload_libraries.

4. Select PGAUDIT from the dropdown menu.
5. Click on Save.
6. Restart the server to apply the change.
7. Connect to your Azure Database for PostgreSQL server as an admin user by using a compatible client like psql.
8. Check that pgaudit is loaded in shared_preload_libraries by executing the following query:
You should see pgaudit in the results.

```
postgres=> show shared_preload_libraries;
              shared_preload_libraries
-----
pg_stat_statements, eventlogger, pg_qs, pgms_wait_sampling, pgaudit
(1 row)
```

9. Enable the pgAudit extension:

```
CREATE EXTENSION pgaudit;
```

Setting the server parameters

For the definition of each parameter, please refer to the [pgaudit documentation](#).

1. Select your Azure Database for PostgreSQL server.
2. On the left, under Settings, select Server parameters.
3. Search for the following parameters and assign them the values listed below:

Parameter	Value	Notes
log_connections	ON	
log_disconnections	ON	
log_duration	ON	
log_statement	ALL	
pgaudit.log	ALL	Setting the pgaudit.log parameter

		to all will log all events. To set up a more granular Audit Policy, see pgAudit documentation .
log_line_prefix	<pre> SONAR_AUDIT=1 TIMESTAMP= %m APPLICATION_NAME=%a U SER=%u DATABASE=%d REMOT E_HOST_AND_PORT=%r SQL_S TATE=%e SESSION_ID=%c SE SSION_START=%s PROCESS_I D=[%p] VIRTUAL_TRANSACTI ON_ID=%v TRANSACTION_ID= %x </pre>	

Note:

Please keep the log_line_prefix in the same format as shown above

For more information, please refer to the [PostgreSQL runtime config logging](#) documentation.

4. Select Save to save your changes.
5. Restart your Azure Database for PostgreSQL server for the changes to take effect.

Stream the logs to an Event Hub

1. Under the Monitoring section in the sidebar, select Diagnostic settings.
2. Click on "+ Add diagnostic setting".
3. Provide a diagnostic setting name.
4. Click on "Stream to an event hub" check box and choose your Subscription, event hub namespace and event hub name from the drop-down.
For "Event hub policy name", use RootManagedSharedAccessKey.
5. Select "PostgreSQL Server Logs" under the categories section.
6. Select Save.

For more information on configuring Diagnostic settings using the Azure Portal, see [Set up Diagnostic logs](#).

Via Azure CLI

1. Run the following Azure CLI command to set the shared_preload_libraries parameter to pgaudit:

```
az postgres server configuration set --name shared_preload_libraries --resource-group <myresourcegroup>
```

- Restart the Azure Database for PostgreSQL server through the cli to apply the change.
- Connect to your Azure Database for PostgreSQL server as an admin user by using a compatible client like psql.
- Check that pgaudit is loaded in shared_preload_libraries by executing the following query:
You should see pgaudit in the query result that will return shared_preload_libraries.

```
postgres=> show shared_preload_libraries;
              shared_preload_libraries
-----
pg_stat_statements, eventlogger, pg_qs, pgms_wait_sampling, pgaudit
(1 row)
```

- Connect to your Azure Database for PostgreSQL server by using a client like psql, and enable the pgaudit extension:

```
CREATE EXTENSION pgaudit;
```

Setting the server parameters

On the Azure CLI run the following commands to set the logging parameters. For the definition of each parameter, please refer to the [pgaudit documentation](#).

- Set the log_connections parameter to ON:

```
az postgres server configuration set --name log_connections --resource-group <myresourcegroup>
```

- Set the log_disconnections parameter to ON:

```
az postgres server configuration set --name log_disconnections --resource-group <myresourcegroup>
```

- Set the log_duration parameter to ON:

```
az postgres server configuration set --name log_duration --resource-group <myresourcegroup>
```

- Set the log_statement parameter to ALL:

```
az postgres server configuration set --name log_statement --resource-group <myresourcegroup>
```

- Set the `pgaudit.log` parameter to ALL:

```
az postgres server configuration set --name pgaudit.log --resource-group <myresourcegroup>
```

*Setting the **pgaudit.log** parameter to **all** will log all events. To set up a more granular Audit Policy, see [Using pgAudit to log database activity](#).

- Set the `log_line_prefix` parameter to the following value:

```
az postgres server configuration set --name log_line_prefix --resource-group <myresourcegroup>
```

Note:

Please keep the `log_line_prefix` in the same format as shown above.

For more information, please refer to the [PostgreSQL runtime config logging](#) documentation.

- Restart the Azure Database for PostgreSQL server for the changes to take effect.

Stream the logs to an Event Hub

Replace the values in the resource, event-hub and event-hub-rule parameters below to create a diagnostic setting.

```
az monitor diagnostic-settings create \
--name diagnosticname \
--resource /subscriptions/<xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx>/resourceGroups/<myresourcegroup> \
--logs '[{"category": "PostgreSQLLogs", "enabled": true}]' \
--event-hub /subscriptions/<xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx>/resourceGroups/<myresourcegroup> \
--event-hub-rule /subscriptions/<xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx>/resourceGroups/<myresourcegroup>
```

For more information refer to Azure product documentation [Azure CLI diagnostic-settings](#).

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet

-
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Asset Specifications](#)

[Azure PostgreSQL Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.

3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

- Asset template name for Azure Database for PostgreSQL Single Server: **AZURE_POSTGRES_SQL_template.xlsx**
- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template name for Azure account; choose ONE of the 3 templates listed below based on the type of authentication mechanism:

- **AZURE_MANAGED_IDENTITY_template.xlsx**
- **AZURE_AUTH_FILE_template.xlsx**
- **AZURE_CLIENT_SECRET_template.xlsx**

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

1. Gateway service log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/postgresql_eventhub.log
```

2. Run the following commands to verify the status of the gateway services:

```
systemctl status -l sonareventhub
systemctl status -l sonareventhub.timer
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the

USC.

- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure Database for PostgreSQL Flexible Server Onboarding Steps

This topic reviews the steps required to audit activity on Azure Database for PostgreSQL Flexible Server and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Audit Data Scope

The scope of audit data collection varies between data sources. Prior to enabling auditing, read through the auditing scope for this data source to identify any potential benefits or constraints. For more information on data collections, see [Main Collections in sonargd Database](#).

- This source collects audit events in the **instance**, **exception**, and **session** collections. Optionally, events can also be collected in the **slow_query** collection.
- When the source generates multiple log lines for a single action, it uses metrics to aggregate them into a single document and stores it in the instance, exception, or session collection based on the event type.
- An exception occurs when a syntactically correct query fails during execution. In this case, two log events may be generated with different classifications: one successful `LOG: 00000: AUDIT:` event and one failed `ERROR/DETAIL/STATEMENT` event. Because they differ in classification and no transaction or query ID is available to correlate them, they cannot be merged and are stored as two separate documents.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps.

Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- If importing manually: You have access to the Event Hub and storage account secret keys for pulling the audit logs.
- If importing via Discovery: A client secret, auth_file or managed identity is needed for DSF to access Azure resources.
- You have permissions to create diagnostic settings in your Azure Database for PostgreSQL to configure audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the database.
- You have a Container within a Storage account for DSF metadata storage.

Authentication Mechanisms for Discovery with Azure Cloud Account Asset

DSF supports the following authentication mechanisms for accessing Azure resources:

- Client secret: for detailed instructions on creating a client secret, please see [Register a Microsoft Entra app and create a service principal](#) and [Option 3: Create a new client secret](#).
- Managed Identity: for detailed instructions on managed identities, please see [Configure managed identities on Azure virtual machines \(VMs\)](#).
- Auth file: a JSON file that contains the client secret credentials (see an example in the [Azure Asset Specifications](#)). Place this file in the \$JSONAR_LOCALDIR/credentials directory on the Agentless Gateway, and ensure that its absolute path is specified in the “key_file” field of the Azure cloud account connection document.

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

```
Microsoft.Authorization/permissions/read
Microsoft.EventHub/namespaces/authorizationrules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/write
Microsoft.EventHub/namespaces/eventhubs/read
Microsoft.EventHub/namespaces/eventhubs/write
Microsoft.EventHub/namespaces/read
Microsoft.EventHub/namespaces/write
Microsoft.Storage/storageAccounts/listkeys/action
```

```
Microsoft.Storage/storageAccounts/read
Microsoft.Storage/storageAccounts/write
Microsoft.Insights/DiagnosticSettings/Read
Microsoft.Insights/DiagnosticSettings/Write
Microsoft.DBforPostgreSQL/flexibleServers/read
```

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

-
- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
 - For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: `<name-of-eventhub-namespace>.servicebus.windows.net`
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

Database Prerequisites

Below are the requirements for granting database permissions for audit log retrieval. Please ensure these are completed by a Database Administrator.

- You have a General Purpose or Memory Optimized Azure PostgreSQL Server. Only these types of Azure Database for PostgreSQL Servers support enabling of the pgAudit extension, which is a prerequisite to collect audit.
- You can connect to your Azure Database for PostgreSQL Server, with an admin user that also has "pg_read_all_settings" permissions, for creating and verifying the creation of the pgaudit extension.
- A restart of the database will be required for the pgaudit settings to take into effect.

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

This documentation provides instructions to enable audit using either "Azure Portal" or "CLI". Please select either method and refer to the [Audit logging in Azure Database for PostgreSQL - Flexible Server](#) documentation for more details.

Enable Audit via Azure Portal

1. Installing the pgaudit extension

- Select your Azure Database for PostgreSQL Flexible Server.
- On the left sidebar, under Settings, select **Server parameters**.
- Search for the **azure.extensions** parameter.
- Select **PGAUDIT** as extension you wish to allow-list.
- Also in Server parameters, search for the **shared_preload_libraries** parameter.
- Select **PGAUDIT** from the dropdown menu.
- Click Save.
- Restart the Server to apply the changes.
- Connect to your Azure Database for PostgreSQL Flexible Server as an admin user by using a compatible client like psql.
- Check that pgaudit is loaded in shared_preload_libraries by executing the following query:

You should see pgaudit in the results.

```
postgres=> show shared_preload_libraries;
              shared_preload_libraries
-----
pg_stat_statements, eventlogger, pg_qs, pgms_wait_sampling, pgaudit
(1 row)
```

- Enable the pgAudit extension:

```
CREATE EXTENSION pgaudit;
```

- I. Verify the extension is enabled with the following command.
You should see pgaudit among the results.

```
postgres=> \dx

                                List of installed extensions
  Name      | Version | Schema  | Description
-----+-----+-----+-----
 azure      | 1.0     | pg_catalog | azure extension for PostgreSQL service
 pg_cron     | 1.4-1   | public   | Job scheduler for PostgreSQL
 pgaadauth   | 1.1     | pg_catalog | Azure Active Directory Authentication
 pgaudit     | 1.7     | public   | provides auditing functionality
 plpgsql     | 1.0     | pg_catalog | PL/pgSQL procedural language
(5 rows)
```

2. Setting the Server parameters

For the definition of each parameter, please refer to the [pgaudit documentation](#).

- a. Select your Azure Database for PostgreSQL Flexible Server.
- b. On the left, under Settings, select Server parameters.

Search for the following parameters and assign them the values listed below.

Parameter	Value	Comments
pgaudit.log	all	Setting the pgaudit.log parameter to all will log all events. To set up a more granular Audit Policy, see pgAudit documentation .
log_line_prefix	SONAR_AUDIT=1 TIMESTAMP=	It is required to use the log_line_prefix value exactly as shown. Any modifications may cause system errors.

For more information, please refer to the [PostgreSQL runtime config logging documentation](#).

- c. Click **Save** to save your changes.

3. Enabling Slow Queries Auditing (Optional)

Note:

- Turning on Slow Query Monitoring will not interfere with standard audit collection.
- Queries that exceed the minimum duration threshold will be mapped to a separate collection called `sonargd.slow_query`.
- Make sure the **log_statement** parameter is set to **None**. Any modifications may cause slow query logs to not be ingested.
- This step is **optional** and required only if you choose to audit slow query logs.
- The following setting should be done IN ADDITION to all previous steps.

Slow query logs audit SQL statements that run longer than a specified duration. This duration is set using the Server parameter "log_min_duration_statement", in addition to the Server parameters enabled in the previous step.

- a. To turn on slow query monitoring modify the following Server parameter:

Parameter	Value	Comments
log_min_duration_statement	<value>	where <value> is your desired threshold in milliseconds.

- b. Click **Save** to save your changes.

4. Stream the logs to an Event Hub

- Under the Monitoring section in the sidebar, select Diagnostic settings.
- Click on "+ Add diagnostic setting".
- Provide a diagnostic setting name.
- Select the following log type from the list of categories: **PostgreSQL Server Logs**.
- Click on **Stream to an event hub** check box and choose your Subscription, event hub namespace and event hub name from the drop-down.
For "Event hub policy name", use RootManagedSharedAccessKey.
- Click **Save**.
For more information on configuring Diagnostic settings using the Azure Portal, see [Set up Diagnostic logs](#).

Enable Audit via Azure CLI

1. Installing the pgaudit extension

- a. On the azure cli run the following command to allow list the pgaudit extension:

```
az postgres flexible-server parameter set --resource-group <your resource group> --ser
```

- b. On the azure cli run the following command to set the shared_preload_libraries parameter to pgaudit:


```
az postgres flexible-server parameter set --name shared_preload_libraries --resource-g
```

- c. Restart the Azure Database for PostgreSQL Server through the cli to apply the change.
- d. Connect to your Azure Database for PostgreSQL Server as an admin user by using a compatible client like psql.
- e. Check that pgaudit is loaded in shared_preload_libraries by executing the following query. You should see pgaudit in the query result that will return shared_preload_libraries.

```
postgres=> show shared_preload_libraries;
              shared_preload_libraries
-----
pg_stat_statements, eventlogger, pg_qs, pgms_wait_sampling, pgaudit
(1 row)
```

- f. Connect to your Azure Database for PostgreSQL Server by using a client like psql, and enable the pgaudit extension:

```
CREATE EXTENSION pgaudit;
```

- g. Verify the extension is enabled with the following command. You should see pgaudit in the results.

```
postgres=> \dx
```

List of installed extensions			
Name	Version	Schema	Description
azure	1.0	pg_catalog	azure extension for PostgreSQL service
pg_cron	1.4-1	public	Job scheduler for PostgreSQL
pgaadauth	1.1	pg_catalog	Azure Active Directory Authentication
pgaudit	1.7	public	provides auditing functionality
plpgsql	1.0	pg_catalog	PL/pgSQL procedural language

```
(5 rows)
```

2. Setting the Server parameters

On the Azure CLI run the following commands to set the logging parameters. For the definition of each parameter, please refer to the [pgaudit documentation](#).

- a. Set the **pgaudit.log** parameter to ALL:

```
az postgres flexible-server parameter set --name pgaudit.log --resource-group <myresourcegroup>
```

* Setting the **pgaudit.log** parameter to **all** will log all events. To set up a more granular Audit Policy, see [pgAudit documentation](#).

- b. Set the **log_line_prefix** parameter to the following value:

```
az postgres flexible-server parameter set --name log_line_prefix --resource-group <myresourcegroup>
```

Note:

- It is required to use the log_line_prefix value exactly as shown. Any modifications may cause system errors.

For more information, please refer to the [PostgreSQL runtime config logging documentation](#).

3. Enabling Slow Queries Auditing (Optional)

Note:

- Turning on Slow Query Monitoring will not interfere with standard audit collection.
- Queries that exceed the minimum duration threshold will be mapped to a separate collection called sonargd.slow_query.
- This step is **optional** and required only if you choose to audit slow query logs.
- The following setting should be done IN ADDITION to all previous steps.

Slow query logs audit SQL statements that run longer than a specified duration. This duration is set using the Server parameter "log_min_duration_statement", in addition to the parameters enabled in the previous step.

- a. To turn on slow query monitoring modify the following Server parameter:

```
az postgres flexible-server parameter set --name log_min_duration_statement --resource-group <myresourcegroup>
```

where XXXX is your desired threshold in milliseconds.

4. Stream the logs to an Event Hub

Replace the <> values in resource, event-hub and event-hub-rule below to create a diagnostic setting.

```
az monitor diagnostic-settings create \
--name diagnosticname \
--resource /subscriptions/<xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx>/resourceGroups/<myresourcegroup>
```

```
--logs ' [{"category": "PostgreSQLLogs","enabled": true}] ' \
--event-hub /subscriptions/<xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx>/resourceGroups/<myresou
--event-hub-rule /subscriptions/<xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx>/resourceGroups/<my
```

For more information refer to Azure product documentation [Azure CLI diagnostic-settings](#).

Filtering the azuresu user using null filters (Optional)

The azuresu is an internal user used by Microsoft for maintenance and analytics purposes. It generates a lot of "background noise" in instance and in session collections and cannot be filtered out on the Azure side. You can filter this user out (or any other requested user) by setting up null filters on the Agentless Gateway.

To filter the azuresu user using the sonarw client:

1. On the Agentless Gateway, connect to sonarw and switch to **sonargd** database.

```
source /etc/sysconfig/jsonar
$JSONAR_BASEDIR/bin/mongo localhost:27117/admin -u admin -p <your_password> --quiet
> use sonargd
```

2. Insert the filter into the **postgresql_flexible_eventhub_null_filters_customer** Collection by using the command below. The following command will create the collection if it doesn't already exist.

```
> db.postgresql_flexible_eventhub_null_filters_customer.insertOne({"properties.message" :
```

The above will filter the azuresu user from all collections: instance, exception, session, and slow_query. If you want to filter it on a specific collection, replace the "filter_from" value with an array. For example "filter_from" : ["instance" , "session"]

To filter out other users, replace the value of USER= with your desired user to filter.

3. As root user, restart the sonarrsyslog service for the changes to take effect immediately. (If not, the changes will automatically be updated within 60mins.)

```
sudo systemctl restart sonarrsyslog
```

To filter the azuresu user using the GUI:

1. Create a spreadsheet with the required fields **properties.message** and **filter_from** as column headers, and values **RE(USER=azuresu)** and **all** correspondingly.
2. Upload the spreadsheet. From the DSF Portal main page, go to **Sync Spreadsheet > Sync Spreadsheet**.

- For Database choose: **"sonargd"**.
- For Collection choose: **"postgresql_flexible_eventhub_null_filters_customer"** (Create it if it doesn't already exist).

Please keep the exact mentioned Collection name and do not replace it with any other value.

- For Update option choose **"Insert"** or **"Upsert"**.
- Click on **"Upload"**.

The below will filter the azuresu user from all collections: instance, exception, session. If you want to filter it on a specific collection, replace the "filter_from" value with an array. For example "filter_from" : ["instance", "session"]

To filter out other users, replace the value of USER= with your desired user to filter.

3. As root user, restart the sonarrrsyslog service for the changes to take effect immediately. (If not, the changes will automatically be updated within 60mins.)

```
sudo systemctl restart sonarrrsyslog
```

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Asset Specifications](#)

[Azure PostgreSQL Flexible Server Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the

mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Note:

The Discovery feature is supported in DSF from version 4.16 onwards.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.

3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

To import the Azure Database for PostgreSQL Flexible Server asset, two assets are required: an Azure PostgreSQL Flexible Server asset and an Azure Event Hub asset.

- Asset template name: **AZURE_POSTGRESQL_FLEXIBLE.template.xlsx**

In the spreadsheet template, you will need to provide the following details for the Event Hub asset:

- **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
- **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
- **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template name for Azure account; choose ONE of the three templates listed below based on the type of authentication mechanism:

- **AZURE_MANAGED_IDENTITY_template.xlsx**
- **AZURE_AUTH_FILE_template.xlsx**
- **AZURE_CLIENT_SECRET_template.xlsx**

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

1. Gateway service log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/postgresql_flexible_eventhub.log
```

2. Run the following commands to verify the status of the gateway services:

```
systemctl status -l sonareventhub
systemctl status -l sonareventhub.timer
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Atlassian

Azure Databricks Onboarding Steps

This topic reviews the steps required to audit activity on Azure Databricks and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The

following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- Your Azure Databricks service is on the Premium pricing tier to enable audit logging.
- You have Workspace Admin permissions to enable the Verbose Audit Logging feature on the Azure Databricks Workspace.
- You have permissions to create diagnostic settings in your Azure Databricks service to configure audit.
- You have an Event Hub within an Event Hub Namespace for receiving audit logs from the data source.
- You have a Container within a Storage account for DSF metadata storage.
- You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is

not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.

- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID=<user assigned identity client id>
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

This documentation provides instructions for enabling audit using either the **Azure Portal** or **Azure CLI**. Please choose one of the methods.

Note:

As of version 4.16, DSF supports the auditing of diagnostic logs from the "Databricks Accounts", "Databricks SQLPermissions" and "Databricks DatabricksSQL" categories only. For more information, see [Diagnostic log reference](#).

Enable Auditing using Azure Portal

To configure audit logging, please follow the below steps.

1. Sign in to the Azure portal.
2. Select your Azure Databricks Workspace.

Set up diagnostic setting

1. Under the Monitoring section in the sidebar, select Diagnostic settings.
2. Select "Add diagnostic setting".
3. Provide a diagnostic setting name.
4. Under the "Destination details" section, click on the "Stream to an event hub" checkbox.
5. From the "Event hub namespace" dropdown, choose your Event Hub namespace.
6. From the "Event hub name (optional)" dropdown, select the Event Hub created in Step 1 "Gathering Permissions & Information" in your chosen Event Hub namespace.

Note:

It is not recommended to choose the "(Create in selected namespace)" dropdown option, which creates one Event Hub per selected log category in the chosen Event Hub namespace. If this option is selected, in Step 3 "Collecting Audit Data", it is required to create an Event Hub asset with a unique Azure Storage Account Container per auto-generated Event Hub to audit all diagnostic log data.

Selecting an individual Event Hub in which to stream the diagnostic logs allows the use of only one Azure Storage Account Container and the import of only one Event Hub asset in Step 3.

7. From the "Event hub policy name" dropdown, select "RootManageSharedAccessPolicy".
8. Under the "Logs" section to the left, select the following categories:
 - a. Databricks Accounts
 - b. Databricks SQLPermissions
 - c. Databricks DatabricksSQL
9. Select Save.

For more information on configuring Diagnostic settings using the Azure Portal, see [Configure diagnostic log delivery](#).

Enable Auditing using Azure CLI

Set up diagnostic logs

Use the following command to setup a diagnostic setting to stream audit logs to an Event Hub. In the command below, specify the *diagnostic-name*, *subscription-id*, *resource-group*, *workspace*, *eventhub-namespace* and *eventhub* values. All the other values should remain the same.

```
az monitor diagnostic-settings create --name <diagnostic-name> \
  --resource /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.Databricks/<workspace-id> \
  --logs '[{"category": "accounts", "enabled": true}, {"category": "databrickssql", "enabled": true}]' \
  --event-hub /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.EventGrid/<event-hub-name> \
  --event-hub-rule /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.EventGrid/<event-hub-name>/rules/<rule-name>
```

For more information, see the Azure Databricks documentation: [az monitor diagnostic-settings create](#).

Enable Verbose Audit logs

The Verbose Audit Logs feature generates additional logs that allow DSF to capture the SQL text of statements that are submitted to Databricks workspaces. Follow the below steps to enable Verbose Audit Logs:

1. As a workspace admin, navigate to the Azure Databricks [admin settings page](#).
2. Click the **Advanced** tab.
3. Under the "Other" section, click on the toggle beside **Verbose Audit Logs** to enable the feature.

For more information on Verbose Audit Logs, see the Azure Databricks documentation: [Verbose audit log events](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Databricks Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

- Asset template name for Azure Databricks Workspace - **AZURE_Databricks_Workspace_template.xlsx**
- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking →

Access keys → key1.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s), please review the following:

1. Gateway service log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/databricks_workspace_eventhub.log
```

2. Run the following commands to verify the status of the gateway services:

```
sudo systemctl status -l sonareventhub
sudo systemctl status -l sonareventhub.timer
sudo systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure Data Explorer Onboarding Steps

This topic reviews the steps required to audit activity on Azure Blob Storage and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Azure Data Explorer

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✔STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- You have permissions to create diagnostic settings in your Azure Data Explorer cluster to configure audit.
- You have an Event Hub within an Event Hub Namespace for receiving audit logs from the data source.
- You have a Container within a Storage account for DSF metadata storage.
- You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of

AZURE_CLIENT_ID to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID=<user assigned identity client id>
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

This documentation provides instructions for enabling audit using either the **Azure Portal** or **Azure CLI**. Please choose one of the methods.

Enable Auditing using Azure Portal

To configure audit logging, please follow the below steps.

1. Sign in to the Azure portal.

2. Select your Azure Data Explorer cluster.

Set up diagnostic logs

1. Under the "Monitoring" section in the sidebar, select "Diagnostic settings".
2. Click on "Add diagnostic setting".
3. Provide a diagnostic setting name.
4. Under the "Logs" section to the left, select the following categories:
 - a. Succeeded ingestion
 - b. Failed ingestion
 - c. Command
 - d. Query
5. Under the "Destination details" section, click on "Stream to an event hub" checkbox.
6. From the "Event hub namespace" dropdown, choose your Event Hub namespace.
7. From the "Event hub name (optional)" dropdown, select the Event Hub created in Step 1 "Gathering Permissions & Information" in your chosen Event Hub namespace.

Note:

It is not recommended to choose the "(Create in selected namespace)" dropdown option which will create one Event Hub per selected log category in the chosen Event Hub namespace. If you choose to do so, when importing the assets in Step 3 "Collecting Audit Data", create one Event Hub asset with a unique Azure Storage Account Container per auto-generated Event Hub to audit all diagnostic log data.

Selecting an individual Event Hub to which to stream the diagnostic logs will allow the use of only one Azure Storage Account Container and the import of only one Event Hub asset in Step 3.

8. From the "Event hub policy name" dropdown, select "RootManageSharedAccessPolicy".
9. Select Save.

For more information on configuring Diagnostic settings using the Azure Portal, see [Configure diagnostic log delivery](#).

Enable Auditing using Azure CLI

Set up diagnostic logs

Use the following command to setup a diagnostic setting to stream audit logs to an Event Hub. In the command below, specify the *diagnostic-name*, *subscription-id*, *resource-group*, *workspace*, *eventhub-namespace* and *eventhub* values. All the other values should remain the same.

```
az monitor diagnostic-settings create --name <diagnostic-name> \
  --resource /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.Insights
  --logs '[{"category": "Succeeded ingestion", "enabled": true}, {"category": "Failed ingestion", "enabled": true}]' \
  --event-hub /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.EventHub
  --event-hub-rule /subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.EventHub
```

For more information, see the Azure documentation: [az monitor diagnostic-settings create](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Data Explorer Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Note:

To monitor slow queries, enter an integer value in the **Duration Threshold** field to specify the execution time in milliseconds after which a query is flagged as slow.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

- Asset template spreadsheet name for Azure Data Explorer: **AZURE_DATA_EXPLORER_template.xlsx**
- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.
- To monitor slow queries, add the **duration_threshold** column to the AZURE DATA EXPLORER asset row and set an integer value for the query execution time in milliseconds. Queries that exceed this threshold will be flagged as slow.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s), please review the following:

1. Gateway service log files:

```
$JSONAR_LOGDIR/eventhub/eventhub.log
$JSONAR_LOGDIR/gateway/syslog/data_explorer_eventhub.log
```

2. Run the following commands to verify the status of the gateway services:

```
sudo systemctl status -l sonareventhub
sudo systemctl status -l sonareventhub.timer
sudo systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Atlassian

Azure Data Lake Storage Gen2 Onboarding Steps

This topic reviews the steps required to audit activity on Azure Data Lake Storage databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- You have access to the Event Hub and Storage Account secret keys for pulling the audit logs.
- You have permissions to create diagnostic settings for your Azure Data Lake Storage Gen2 to configure audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the database.
- You have a Container within a Storage Account for DSF metadata storage.

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is

not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.

- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers

and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Specify the networks permitted to access the account. This can be done from the Advanced tab when creating the service, or navigating to the Settings menu on your storage account. See [Azure documentation](#) for more details.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

1. In the Azure portal select Storage accounts, then the name of the storage account to open the storage account blade.
2. Select Diagnostic settings (classic) in the Monitoring (classic) section of the menu blade.
 - a. Select the type of metrics data for each service you wish to monitor, and the retention policy for the data. For more details see [Azure documentation](#).
 - b. Change the "Status" blade from "Off" to "On", and click "Save".
3. Select Diagnostic settings in the Monitoring section of the menu blade.
 - a. Under "Name" click on each of the resources listed: blob, queue, table, file and add a diagnostic setting for

each of them:

- a. Click on "+ Add diagnostic setting" to add a new diagnostic.
- b. Under "Logs" → "Categories" check the options: StorageRead, StorageWrite, StorageDelete.
- c. Under "Destination details" check the "Stream to an event hub" and from the drop-down list select the corresponding values for: subscription, event hub namespace, event hub name, eventhub policy name (default value is RootManageSharedAccessKey).
- d. Click "Save".

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Data Lake Storage Gen2 Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

- Asset template name for Azure Data Lake Storage Gen2: ***AZURE_DATA_LAKE_STORAGE_GEN2_template.xlsx***
- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

1. Gateway service log files:

```
Sonarrsyslog log file: ${JSONAR_LOGDIR}/gateway/syslog/storage_blob_eventhub.log
Azure Eventhub Service Log file: ${JSONAR_LOGDIR}/eventhub/eventhub.log
```

2. Run the following commands to verify the status of the various services:

```
systemctl status -l sonarrsyslog
systemctl status -l sonareventhub
```

3. After the audit is turned on:

- a. On the Azure portal select the Event Hub and go to the "Messages" dashboard.
- b. Select a time frame period and check the count under the "Outgoing messages" tag. If the count is greater than 0, in the selected time frame, this indicates that messages are being received by the Event Hub.

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure Dedicated SQL Pool Onboarding Steps

This topic reviews the steps required to audit activity on Azure Dedicated SQL data sources and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- If using manual import: you have access to the Event Hub and Storage Account secret keys for pulling the audit logs.
- If using Discovery: you have access to a client secret, auth_file or managed identity for the DSF to access your Azure resources.
- You have an Event Hub within an Event Hub Namespace for receiving the audit logs from the database.
- You have a Container within a Storage Account for DSF metadata storage.

Authentication Mechanisms for Discovery with Azure Cloud Account Asset

DSF supports the following authentication mechanisms for accessing Azure resources:

- Client secret: for detailed instructions on creating a client secret, please see [Register a Microsoft Entra app and create a service principal](#) and [Option 3: Create a new client secret](#).
- Managed Identity: for detailed instructions on managed identities, please see [Configure managed identities on Azure virtual machines \(VMs\)](#).

- Auth file: a JSON file that contains the client secret credentials (see an example in the [Azure Asset Specifications](#)). Place this file in the \$JSONAR_LOCALDIR/credentials directory on the Agentless Gateway, and ensure that its absolute path is specified in the “key_file” field of the Azure cloud account connection document.

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

```
Microsoft.Authorization/permissions/read
Microsoft.EventHub/namespaces/authorizationrules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/write
Microsoft.EventHub/namespaces/eventhubs/read
Microsoft.EventHub/namespaces/eventhubs/write
Microsoft.EventHub/namespaces/read
Microsoft.EventHub/namespaces/write
Microsoft.Insights/DiagnosticSettings/Read
Microsoft.Insights/DiagnosticSettings/Write
Microsoft.Sql/servers/read
Microsoft.Storage/storageAccounts/blobServices/containers/read
Microsoft.Storage/storageAccounts/blobServices/containers/write
Microsoft.Storage/storageAccounts/listkeys/action
Microsoft.Storage/storageAccounts/read
Microsoft.Storage/storageAccounts/write
```

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To

create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Note:

All the resources must be created under the same resource group and the same region as the Azure Dedicated SQL Pool.

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Please **choose one** of the following methods to enable audit logging:

1. Enable Audit via DSF
2. Enable Audit via Azure Portal

Enable Audit via DSF

Server audit can be enabled via DSF by running "Connect Gateway" in Asset Dashboard or "Enable Audit Collection" in USC. No further action is required here, please skip to Step 3 "Collecting Audit Data".

Note:

The default Azure policy is enabled on the server, for more information please see [Configure Auditing for your server](#).

Enable Audit via Azure Portal

Alternatively, to enable server-level auditing manually, please follow the below steps.

Note:

Enabling audit on the server-level will audit all SQL pools in the SQL Server.

1. Open the Azure portal.
2. Navigate to the Azure SQL Server that has the Dedicated SQL pool.
3. Navigate to **Auditing** under the **Security** heading in the lefthand pane.
4. Toggle "Enable Azure SQL Auditing" to enable auditing.
5. Choose "Event Hub" as the audit log destination and select the Event Hub that you created as part of Step 1.
6. Click on "Save".

By doing this, Azure enables the default auditing policy which will audit all the queries and stored procedures executed against the database, as well as successful and failed logins. For more details, see [Configure Auditing for your server](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

Note that Azure Dedicated SQL Pool (formerly SQL DW) and Azure SQL Server share the same Asset Specifications and asset template spreadsheet.

[Azure Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

[Azure Dedicated SQL pool \(formerly SQL DW\) Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.

2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.

3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

To import the Azure Dedicated SQL Pool asset, two assets are required: an Azure SQL Server asset and an Azure EventHub asset.

- Asset template name: **AZURE_SQL_SERVER_template.xlsx**

In the spreadsheet template, you will need to provide the following details for the Event Hub asset:

- **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
- **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
- **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template name for Azure account - choose ONE of the templates listed below based on the type of authentication mechanism:

- **AZURE_MANAGED_IDENTITY_template.xlsx**
- **AZURE_AUTH_FILE_template.xlsx**
- **AZURE_CLIENT_SECRET_template.xlsx**

The connection documents of discovered assets will not include some fields like "username", "password" etc. If you require these fields to run playbooks like "Test Connection", "Turn On Audit Policy", update the database connection document to include all the relevant fields for your Data Source. To do this go to "Connection Dashboard" → Filter for the document and click on the update document button located at the right end.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

1. Gateway service log files:

```
Sonarrsyslog service: $JSONAR_LOGDIR/gateway/syslog/mssql_eventhub.log
Azure Event Hub service: $JSONAR_LOGDIR/eventhub/eventhub.log
```

2. Run the following commands to verify the status of the various services:

```
systemctl status -l sonarrsyslog
systemctl status -l sonareventhub
```

3. After the audit policy is turned on and the database starts generating traffic:

- a. On the Azure portal, select your Event Hub and go to the "Messages" dashboard. Select a time frame period and check the count under the "Outgoing messages" tag. If the count is greater than 0 in the selected time frame, this indicates that messages are being received by the Event Hub.
- b. To verify that the policy is enabled on your server, run the following command, filling in the values for <resource-group> and <sql-server-name>:

```
az sql server audit-policy show --resource-group <resource-group> --name <sql-server-name>
```

For more information, see [az sql server audit-policy](#).

- c. On the Agentless Gateway host(s), check the \$JSONAR_DATADIR/eventhub/<eventhub_name> folder to see that *.json files are being received.

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

-
- [Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.](#)
 - [Adding assets via Unified Settings Console \(USC\):](#) Instructions to add and edit data source assets via the USC.
 - [Adding Assets via the Import Assets Page:](#) Instructions to add data source assets via the Asset Dashboard.
 - [Using the Data Security Fabric Portal:](#) An overview of the Data Security Fabric (DSF) portal.
 - [Data Security Coverage Tool:](#) Systems and version compatibility with DSF Hub.

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Azure SQL Managed Instance Onboarding Steps

This topic reviews the steps required to audit activity on Azure SQL Managed Instance and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- If onboarding the assets manually via the Asset Dashboard or USC: You have access to the Event hub and Storage Account secret keys for pulling the audit logs.
- If onboarding the assets using the Discovery feature: You have a client secret, auth_file or managed identity in order for DSF to access Azure resources.
- You have permissions to create diagnostic settings in your SQL Managed Instance Account for configuring audit.
- You have an Event Hub within an Event Hub Namespace to receive the audit logs from the data source.
- You have a Container within a Storage Account for DSF metadata storage.

Authentication Mechanisms for Discovery with Azure Cloud Account Asset

DSF supports the following authentication mechanisms for accessing Azure resources:

- Client secret: for detailed instructions on creating a client secret, please see [Register a Microsoft Entra app and create a service principal](#) and [Option 3: Create a new client secret](#).
- Managed Identity: for detailed instructions on managed identities, please see [Configure managed identities on Azure virtual machines \(VMs\)](#).
- Auth file: a JSON file that contains the client secret credentials (see an example in the [Azure Asset Specifications](#)). Place this file in the \$JSONAR_LOCALDIR/credentials directory on the Agentless Gateway, and ensure that its absolute path is specified in the "key_file" field of the Azure cloud account connection document.

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

```
{
  "permissions": [
    {
      "actions": [
        "Microsoft.Authorization/permissions/read",
        "Microsoft.EventHub/namespaces/authorizationrules/listkeys/action",
        "Microsoft.EventHub/namespaces/eventhubs/authorizationRules/listkeys/action",
        "Microsoft.EventHub/namespaces/eventhubs/authorizationRules/write",
        "Microsoft.EventHub/namespaces/eventhubs/read",
        "Microsoft.EventHub/namespaces/eventhubs/write",
        "Microsoft.EventHub/namespaces/read",
        "Microsoft.EventHub/namespaces/write",
        "Microsoft.Insights/DiagnosticSettings/Read",
        "Microsoft.Insights/DiagnosticSettings/Write",
        "Microsoft.Sql/servers/auditingSettings/read",
        "Microsoft.Sql/servers/auditingSettings/write",
        "Microsoft.Sql/servers/databases/auditingSettings/read",
        "Microsoft.Sql/servers/databases/auditingSettings/write",
        "Microsoft.Sql/servers/databases/extendedAuditingSettings/read",
        "Microsoft.Sql/servers/databases/extendedAuditingSettings/write",
        "Microsoft.Sql/servers/extendedAuditingSettings/read",
        "Microsoft.Sql/servers/extendedAuditingSettings/write",
        "Microsoft.Sql/servers/read",
        "Microsoft.Storage/storageAccounts/blobServices/containers/read",
        "Microsoft.Storage/storageAccounts/blobServices/containers/write",
        "Microsoft.Storage/storageAccounts/listkeys/action",
        "Microsoft.Storage/storageAccounts/read",
        "Microsoft.Storage/storageAccounts/write",
        "Microsoft.Sql/managedInstances/read"
      ]
    }
  ]
}
```

```
"notActions": [],
"dataActions": [],
"notDataActions": [] }
}] }
```

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:

When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.

- For DSF version 4.12:

Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

- Enable a public endpoint for your SQL Managed instance.
- Configure your SQL Managed instance network security group to allow traffic to the SQL Managed instance public endpoint. For more details, please see [Configure public endpoints in Azure SQL Managed Instance](#) to create a new server principal for the purpose of creating the Server Audit, the following

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: `<name-of-eventhub-namespace>.servicebus.windows.net`
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.

- Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.

Database Prerequisites

Below are the requirements for granting database permissions for audit log retrieval. Please ensure these are completed by a Database Administrator.

- You need a server principal with the following privileges in order to create a [Server Audit](#) and [Server Audit Specification](#).
 - ALTER ANY SERVER AUDIT: to create, alter, or drop a Server Audit and a Server Audit Specification.
- If you creag privilege is also required for that principal:..
 - CONTROL SERVER: to view a Server Audit Specification after it has been created. Server Audit Specifications are only viewable by the sysadmin account, principals with explicit access to the Server Audit, and users with the CONTROL SERVER privilege.
- VIEW ANY DEFINITION ?

For more details on the required privileges for auditing Azure SQL Managed instances, see [Permissions](#)

Create Audit Management User (Optional)

This step creates a new database user with the privileges necessary for creating and managing audit policies. This user will also require the following privileges:

- ALTER ANY SERVER AUDIT: to create, alter, or drop a Server Audit and a Server Audit Specification.
- CONTROL SERVER: to view a Server Audit Specification after it has been created.

1. To create a new user, connect to your SQL Managed Instance and run the following commands:

```
CREATE LOGIN <login_username> WITH PASSWORD=<your_password>;
GO

CREATE USER <login_username> FROM LOGIN <login_username>;
GO
```

2. Grant the following permissions to the server principle created above.

```
GRANT ALTER ANY SERVER AUDIT TO <login_username>;
GO

GRANT CONTROL SERVER TO <login_username>;
GO
```

Validate Connection to the Database

1. To verify the connection to the SQL Managed Instance is error-free, login as the "sonarw" user on the Agentless Gateway host:

```
sudo su - sonarw
```

2. Source the DSF variables:

```
source /etc/sysconfig/jsonar
```

3. Connect to the database:

```
${JSONAR_BASEDIR}/bin/tsql -H <SQL Managed Instance endpoint> -p <port> -U <login_username>
```

4. If everything is configured correctly, you should be able to connect and run a simple query like below:

```
1> select db_name();
2> GO

master
(1 row affected)
```

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Configure Diagnostic Logging

To configure database auditing that will send the audit data to the Event hub created in the previous step, complete the following steps:

1. Navigate to the managed instance in the Azure portal.
2. Under the Monitoring section in the sidebar, select Diagnostic settings.
3. Select "Add diagnostic setting".
4. Provide a Diagnostic Setting Name.
5. Under the "Logs" categories section, select the "SQL Security Audit Event" category.
6. Under the "Destination details" section, click on the "Stream to an event hub" checkbox.
7. From the "Event hub namespace" dropdown, choose your Event Hub namespace.

8. From the "Event hub name (optional)" dropdown, select the Event Hub created in Step 1 "Gathering Permissions & Information" in your chosen Event Hub namespace.
9. From the "Event hub policy name" dropdown, select "RootManageSharedAccessPolicy".
10. Select Save.

For more details, see the [Set up auditing for your server to Event Hubs or Azure Monitor logs](#).

Create a Server Audit

To create and set up a server audit, complete the following steps:

1. Connect to the SQL Managed instance using a user with the privileges outlined in "Step 1 Onboarding Prerequisites" and a supported client. This user may be the optional Audit Management user, or another user.
2. Create and enable a server audit by running the below commands on master database, providing a name and filling in the value for <server audit>.

```
USE master;
GO

CREATE SERVER AUDIT <server audit> TO EXTERNAL_MONITOR;
GO

ALTER SERVER AUDIT <server audit> WITH (STATE=ON);
GO
```

3. Create a server audit specification or database audit specification for server audit <server audit>.

```
CREATE SERVER AUDIT SPECIFICATION <server_audit_specification> FOR SERVER AUDIT <server audit>;
GO
```

4. Ensure that your server audit policy is active. Check the state and audit configuration with the following commands:

```
SELECT * FROM sys.dm_server_audit_status;
GO
```

5. Check the audit specification with the following command:

```
SELECT * FROM sys.server_audit_specifications;
GO
```

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Asset Specifications](#)

[Azure SQL Managed Instance Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

Asset template spreadsheet name for Azure SQL Managed Instance: **AZURE_SQL_MANAGED_INSTANCE_template.xlsx**

- In the spreadsheet template, you will need to provide the following details for the Event Hub asset:
 - **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
 - **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
 - **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template name for Azure account; choose ONE of the templates listed below based on the type of authentication mechanism:

- **AZURE_MANAGED_IDENTITY_template.xlsx**
- **AZURE_AUTH_FILE_template.xlsx**
- **AZURE_CLIENT_SECRET_template.xlsx**

The connection documents of discovered assets will not include some fields like "username", "password" etc. If you require these fields to run playbooks like "Test Connection", update the database connection document to include all the relevant fields for the data source. To do this, navigate to the "Connection Dashboard" → Filter for the document → click on the Update document button located at the right end.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s), please review the following:

1. Gateway service log files:
 - a. Sonarrrsyslog service: `$JSONAR_LOGDIR/gateway/syslog/mssql_managed_instance_eventhub.log`
 - b. Azure Event Hub service: `$JSONAR_LOGDIR/eventhub/eventhub.log`
2. Run the following commands to verify the status of the various services:

```
systemctl status -l sonarrrsyslog
systemctl status -l sonareventhub
```

3. After the audit policy is turned on and the database starts generating traffic:
 - a. On the Azure portal, select your Event Hub and go to the "Messages" dashboard. Select a timeframe period and check the count under the "Outgoing messages" tag. If the count is greater than 0 in the selected timeframe, this indicates that messages are being received by the Event Hub.
 - b. On the Agentless Gateway host(s), check the `$JSONAR_DATADIR/eventhub/<your_eventhub_name>` folder to see that *.json files are being received.

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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Azure SQL Server Onboarding Steps

This topic reviews the steps required to audit activity on Azure SQL Server and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Audit Data Scope

The scope of audit data collection varies between data sources. Prior to enabling auditing, read through the auditing scope for this data source to identify any potential benefits or constraints. For more information on data collections, see [Main Collections in sonargd Database](#).

- This source collects audit events in the **instance**, **exception**, and **session** collections.
- In the **instance** collection, the `Original SQL` field contains the executed SQL query. The `Objects & Verbs` field is extracted from that query, except for stored procedure actions, where `Objects & Verbs` is manually concatenated with `Object` set to `N/A` and `Verb` set to the stored procedure name.
- `Client Port` is not included in audit data. The field defaults to `N/A`.

-
- The following events are not audited by this source:
 - `LOGOFF` for non-contained database users. For more information, see [Make your database portable by using contained databases](#). When using the Azure default policy, `LOGOFF` is not audited at all because the `DATABASE_LOGOUT_GROUP` audit option is not included in the default policy.
 - `FAILED LOGON` due to an incorrect database name or port value.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

- If using manual import: you have access to the Event Hub and Storage Account secret keys for pulling the audit logs.
- If using Discovery: you have access to a client secret, `auth_file` or managed identity for the DSF to access your Azure resources.
- You have permissions to create diagnostic settings in the Azure SQL Server to configure audit.
- You have an Event Hub within an Event Hub Namespace for receiving the audit logs from the database.
- You have a Container within a Storage Account for DSF metadata storage.

Authentication Mechanisms for Discovery with Azure Cloud Account Asset

DSF supports the following authentication mechanisms for accessing Azure resources:

- Client secret: for detailed instructions on creating a client secret, please see [Register a Microsoft Entra app and create a service principal](#) and [Option 3: Create a new client secret](#).
- Managed Identity: for detailed instructions on managed identities, please see [Configure managed identities on Azure virtual machines \(VMs\)](#).
- Auth file: a JSON file that contains the client secret credentials (see an example in the [Azure Asset Specifications](#)). Place this file in the `$JSONAR_LOCALDIR/credentials` directory on the Agentless Gateway, and ensure that its absolute path is specified in the “`key_file`” field of the Azure cloud account connection

document.

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

```
Microsoft.Authorization/permissions/read
Microsoft.EventHub/namespaces/authorizationrules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/listkeys/action
Microsoft.EventHub/namespaces/eventhubs/authorizationRules/write
Microsoft.EventHub/namespaces/eventhubs/read
Microsoft.EventHub/namespaces/eventhubs/write
Microsoft.EventHub/namespaces/read
Microsoft.EventHub/namespaces/write
Microsoft.Insights/DiagnosticSettings/Read
Microsoft.Insights/DiagnosticSettings/Write
Microsoft.Sql/servers/read
Microsoft.Storage/storageAccounts/blobServices/containers/read
Microsoft.Storage/storageAccounts/blobServices/containers/write
Microsoft.Storage/storageAccounts/listkeys/action
Microsoft.Storage/storageAccounts/read
Microsoft.Storage/storageAccounts/write
```

To use DSF's Audit Policy Management feature, the following additional privileges should also be granted:

```
Microsoft.Sql/servers/auditingSettings/read
Microsoft.Sql/servers/auditingSettings/write
Microsoft.Sql/servers/databases/auditingSettings/read
Microsoft.Sql/servers/databases/auditingSettings/write
Microsoft.Sql/servers/databases/extendedAuditingSettings/read
Microsoft.Sql/servers/databases/extendedAuditingSettings/write
Microsoft.Sql/servers/extendedAuditingSettings/read
Microsoft.Sql/servers/extendedAuditingSettings/write
```

Creating an Azure Event Hub Namespace and Event Hub

Azure data sources use Azure Event Hubs as the log aggregator, from which DSF Hub retrieves the audit logs. Event Hubs are created and contained within an Event Hub Namespace. To create an Event Hub Namespace and Event Hubs, follow the Azure documentation: [Create an Event Hubs namespace](#) and [Create an Event Hub](#).

Note:

- A single Event Hub can store the audit logs of multiple instances of the same Server Type (data source type e.g. Azure for MariaDB, Azure SQL Server, etc.) Storing the audit logs of multiple Server Types is not supported for audit monitoring by DSF. A Server Type is the type of data source, e.g. Azure Database for MariaDB or Azure SQL Server.
- It is recommended to have one Agentless Gateway pull from a given Azure Event Hub. Multiple Agentless Gateways pulling from the same Event Hub may result in the ingestion of partial data.
- If several instances of the same Server Type write to a single Event Hub, connecting any one of the data source assets or the Event Hub asset will pull all the audit data from the Event Hub. However, by default

the "audit_pulled_enabled" field of newly imported asset documents is set to false, so only data from data source assets with "audit_pull_enabled" set to true will be ingested.

Creating a Storage Account and Container

A Storage Account is a collection of one or more Storage Containers. These Storage Containers store transactional data for the Event Hub import processes, with one container being required for each Event Hub. To create a Storage Account or Storage Container, follow the Azure documentation: [Create a storage account](#).

Note:

All the resources must be created under the same resource group and the same region as the Azure SQL Server Database.

Using Microsoft Entra ID (formerly Azure AD) Authentication Mechanism

Note:

This authentication mechanism is only available for DSF versions 4.12 and newer.

The DSF service will access the Event Hub and Storage Container using either a system or a user-assigned managed identity attached to an Azure Agentless Gateway. By default, the system-assigned managed identity will be used.

To use a user-assigned identity:

- For DSF versions 4.13 and above:
When onboarding the Event Hub asset, add the "user_identity_client_id" field to the connection document with the Client ID of your user-assigned managed identity.
- For DSF version 4.12:
Modify the file `/etc/sysconfig/jsonar_ex` on the Azure gateway with the below content. Modify the value of `AZURE_CLIENT_ID` to the Client ID of your user-assigned managed identity.

```
AZURE_CLIENT_ID="<user assigned identity client id>"
```

The managed identity attached to the Agentless Gateway must be assigned the following roles:

- **Azure Event Hubs Data Receiver:** allows receive access to Azure Event Hubs resources.
- **Storage Blob Data Contributor:** allows for read, write, and delete access to Azure Storage Blob Containers and data.

For more information on managed identities, please see [What are managed identities for Azure resources?](#)

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Microsoft's Event Hub service running on the Azure cloud. Ensure the Agentless Gateway host is able to access the Event Hub Namespace. Note that:

- The Agentless Gateway host must be able to communicate with Microsoft data center hosting the Event Hub from its own data center.
- When configuring the firewall, use the Event Hub Namespace's hostname because it is static, as opposed to the IP of the Event Hub Namespace which is dynamic. The hostname of the Event Hub namespace follows this format: <name-of-eventhub-namespace>.servicebus.windows.net
- Azure Event Hubs uses the AMQPS protocol for event reading, which may require firewall rules to specify it as Allowed.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Event Hub and Storage Account:
 - Ports 5671 and 5672: Azure Event Hubs listens on these ports for AMQP and AMQPS traffic, respectively.
 - Port 443: for HTTPS requests to the Storage Account to maintain the metadata necessary for the pulling of Event Hub data.
- Additionally, ensure the following outbound ports are also open on the Agentless Gateway:
 - Port 1433: to connect to the Azure SQL Server Database for turning on audit policies (required only if using DSF's audit management feature).

Adding the Agentless Gateway IP to Server-level Firewall Rules

To manage access to an Azure SQL Server from a specified IP address or a range of IP addresses server-level firewall rules can be used. Please refer to the [Create and manage IP firewall rules](#) documentation for more details.

Database Prerequisites

Below are the requirements for granting database permissions for audit log retrieval. Please ensure these are completed by a Database Administrator.

Create Audit Management User (optional)

A database user (Audit Management user) is needed for creating and managing audit policies through the DSF

Hub. No extra privileges are required. To create the audit management user, connect to the SQL Server database and run the following commands:

1. Create a login, filling in the value for <login_name> and <password>:

```
CREATE LOGIN <login_name> WITH PASSWORD = '<password>';
GO
```

2. Add a user for the "login" created above, filling in the value for <username> and <password>:

```
CREATE USER <username> FROM LOGIN <login_name>;
GO
```

Note that for ease of use, it is recommended that the login and user are created with the same identifier.

The credentials of this user will be needed in Step 4 "Managing Audit Policies" in a connection document with reason "audit management".

Validate Connection to Database

Verify the connection to the Azure SQL master database is error-free by completing these steps:

1. On the Agentless Gateway host, login as the "sonarw" user:

```
sudo su - sonarw
```

2. Source the DSF variables:

```
source /etc/sysconfig/jsonar
```

3. Connect to the master database, filling in the values for <username>, <password> and <server_name> found in the Overview of the SQL Server in the Azure Portal:

```
$JSONAR_BASEDIR/bin/tsql -S <server_name> -p 1433 -D master -U <username> -P <password>
```

4. Run the select command below. If everything is configured correctly, this will return the list of all the databases under your Azure SQL Server. Address any errors here before continuing.

```
SELECT name FROM sys.databases;
GO
```

✓STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Enable Auditing

Please **choose one** of the following methods to enable audit logging:

1. Enable Auditing via DSF
2. Enable Auditing via the Azure Portal

Enable Auditing via DSF

Server audit can be enabled via DSF by creating audit policies. This will be done in Step 4 "Managing Audit Policies"; after validating your audit policy, please skip ahead to Step 3 "Collecting Audit Data". To see which audit action groups will be enabled by any of the available audit policies, see [Viewing Lists of Actions for Action Groups](#).

Enable Auditing via the Azure Portal

Alternatively, to enable Server level auditing manually, please follow the below steps.

1. Open the Azure portal.
2. Navigate to **Auditing** under the Security heading in your **SQL server** pane.
3. Toggle "Enable Azure SQL Auditing" to enable auditing.
4. Choose "Event Hub" as the audit log destination and select the Event Hub that you created as part of Step 1.
5. Click on "Save".

By doing this, Azure enables the default auditing policy which will audit all the queries and stored procedures executed against the database, as well as successful and failed logins. Note that the default auditing policy includes the following set of action groups:

- BATCH_COMPLETED_GROUP
- SUCCESSFUL_DATABASE_AUTHENTICATION_GROUP
- FAILED_DATABASE_AUTHENTICATION_GROUP

To configure auditing for [different types of actions and action groups](#) using PowerShell, a REST API, or the Azure CLI, see [Manage Azure SQL Database Auditing using APIs](#). For more information, see [Configure Auditing for your server](#).

Validate the Audit Policy

To verify the policy that has been enabled on your server, run the following command, filling in the values for <resource-group> and <sql-server-name>:

```
az sql server audit-policy show --resource-group <resource-group> --name <sql-server-name>
```

Example output of a successful default audit policy displaying only relevant fields:

```
{
  "auditActionsAndGroups": [
    "SUCCESSFUL_DATABASE_AUTHENTICATION_GROUP",
    "FAILED_DATABASE_AUTHENTICATION_GROUP",
    "BATCH_COMPLETED_GROUP"
  ],
  "eventHubAuthorizationRuleId": "/subscriptions/your-subscription-id/resourcegroups/your-r",
  "eventHubName": "your-eventhub-name",
  "eventHubTargetState": "Enabled",
  "id": "/subscriptions/your-subscription-id/resourceGroups/your-resource-group/providers/Mi",
  "name": "Default",
  "resourceGroup": "your-resource-group-name",
  "state": "Enabled",
  "type": "Microsoft.Sql/servers/auditingSettings"
  ...
}
```

For more information, see [az sql server audit-policy](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[Azure Asset Specifications](#)

[Azure Event Hub Asset Specifications](#)

[Azure MS SQL Server Asset Specifications](#)

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click **Advanced**.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

To import the Azure SQL Server asset, two assets are required: an Azure SQL Server asset and an Azure Event Hub asset.

- Asset template name: **AZURE_SQL_SERVER_template.xlsx**

In the spreadsheet template, you will need to provide the following details for the Event Hub asset:

- **eventhub_access_policy**: the name of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by navigating to your namespace → Settings → Shared access policies. The default name used by Azure is "RootManageSharedAccessKey".
- **eventhub_access_key**: the Primary key of the "Shared access policy" for the namespace containing the targeted Event Hub. This can be found by clicking on the Shared access policy found above, and selecting the Primary key.
- **azure_storage_secret_key**: the "Access key" for the storage account holding the dedicated storage container. This can be found by navigating to your storage account → Security + networking → Access keys → key1.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Note:

Asset template name for Azure account; choose ONE of the 3 templates listed below based on your desired authentication mechanism:

- **AZURE_MANAGED_IDENTITY_template.xlsx**
- **AZURE_AUTH_FILE_template.xlsx**
- **AZURE_CLIENT_SECRET_template.xlsx**

The connection documents of discovered assets will not include some fields like "username", "password" etc. If you require these fields to run playbooks like "Test Connection", "Turn On Audit Policy", update the database connection document to include all the relevant fields for your Data Source. To do this go to "Connection Dashboard" → Filter for the document and click on the update document button located at the right end.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

✓STEP 4: Managing Audit Policies

Audit Policies are created to specify which actions will be collected from the database. This can be done in two

ways:

- **Using the Audit Policy Management Feature**
(or)
- **Creating an Audit Policy manually**

Using the Audit Policy Management Feature

DSF provides automated solutions for managing and monitoring audit policies. Audit policies can be managed from the Asset Dashboard (OR) USC.

Managing Audit Policies with Asset Dashboard

To apply an audit policy via Asset Dashboard:

1. From the **DSF Portal**, expand the left-side navigation menu by clicking **>>**. Click **Data Activity Monitoring > Search & Analyze > Dashboard**. Alternatively, you can use the search bar to locate **Dashboard**.
2. On the **Dashboards** page, select the **Asset Details** dashboard.
3. In the **Asset Details** dashboard in DSF DataVis, go to the **Assets** panel and locate the asset to which you want to apply the audit policy.
4. Click the three vertical dots at the end of the row for the asset, in the **Actions** column.
5. In the menu that opens, select **Management > Audit Actions > Turn on Audit Policy**. The **Turn On Audit Policy** window opens.
6. In the **Turn On Audit Policy** window, under **Audit Policy**, select an audit policy by type or by name.
7. Click **Submit**. The audit policy is turned on, that is, applied to the asset.
8. If you want, click **Refresh** to view the asset's audit policy details (e.g. policy_template_name) in the relevant row in the **Assets** panel.

For detailed instructions and supported database versions, see [Managing Audit Policies](#) in the user guide.

Managing Audit Policies with Unified Settings Console

To create and apply a new DSF Hub audit policy via USC:

1. From the main page of **DSF Portal**, under **Apps**, click **Unified Settings Console**.
 2. In the **Appliances** pane, select **DSF Hub**.
 3. Click the **Audit Policies** tab. The audit policies of the DSF Hub appliance are displayed.
 4. Click **Add**.
 5. In the **Configuration** tab, in the **Policy Name** field, type a name for the policy.
-

6. In the **Configuration** tab, select a policy type. Options available in this menu include **Native audit policy**, **Gateway audit policy** and **Combined audit policy**.
7. Select match criteria. Available options depend on audit type selected: For more information about these criteria, see [Supported Criteria for Audit Policies](#).
8. If you selected the **Exclude List** type, add at least one of the following sets:
 - DB User Name
 - Analyzed Client IP
 - Source Program
 You can add multiple sets of **Exclude List** criteria via the **+ Add** button.
9. If you selected the **Action Groups** policy type, add one or more action groups. For more information, see [Viewing Lists of Actions for Action Groups](#).
10. Click the **Applied to** tab and select the data source assets to which the policy will be applied.
11. Turn **Audit collection** on or off. This option provides a convenient method to enable audit collection directly from the audit policy.
12. Click **Save**. DSF Hub begins updating the audit policy; this process may take a few minutes to complete.

For detailed instructions, see [Managing Audit Policies via USC](#).

To use this, it is necessary to create a dedicated audit management user. This user will connect to the database and run queries to create and modify audit policies. Please use the Audit Management user created in Step 1.

Note:

- If the Azure SQL Server also contains a "[Dedicated SQL pool \(formerly SQL DW\)](#)", running the "Turn on Audit Policy" playbook via the DSF is not supported.
- If the DSF's Audit Policy Management feature is being used, all the other existing audit policies at the server level will be removed.
- If you run the "Connect to Gateway" playbook without having previous run "Turn on Audit Policy", the default Azure policy will be enabled on the server. For more information on the default Azure policy, see [Auditing for Azure SQL Database and Azure Synapse Analytics](#).

Validate the Audit Policy

To verify the policy that has been enabled on your server, run the following command, filling in the values for `<resource-group>` and `<sql-server-name>`:

```
az sql server audit-policy show --resource-group <resource-group> --name <sql-server-name>
```

Example output of a successful audit policy displaying only relevant fields:

```
{
  "auditActionsAndGroups": [
    "SUCCESSFUL_DATABASE_AUTHENTICATION_GROUP",
    "FAILED_DATABASE_AUTHENTICATION_GROUP",
    "BATCH_COMPLETED_GROUP"
  ],
  "eventHubAuthorizationRuleId": "/subscriptions/your-subscription-id/resourcegroups/your-resource-group/providers/Microsoft.EventHub/authorizationRules/your-authorization-rule",
  "eventHubName": "your-eventhub-name",
  "eventHubTargetState": "Enabled",
  "id": "/subscriptions/your-subscription-id/resourceGroups/your-resource-group/providers/Microsoft.EventHub/authorizationRules/your-authorization-rule",
  "name": "Default",
  "resourceGroup": "your-resource-group-name",
  "state": "Enabled",
  "type": "Microsoft.Sql/servers/auditingSettings"
  ...
}
```

For more information, see [az sql server audit-policy](#).

Creating an Audit Policy Manually

Alternatively, it is possible to manually create and manage audit policies on the data source itself.

Please see Step 2 "Enabling Audit on the Data Source".

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

1. Gateway service log files:

```
Gateway service: $JSONAR_LOGDIR/gateway/syslog/mssql_eventhub.log
Azure Event Hub service: $JSONAR_LOGDIR/eventhub/eventhub.log
```

2. Run the following commands to verify the status of the various services:

```
systemctl status -l sonarrsyslog
systemctl status -l sonareventhub
systemctl status -l sonareventhub.timer
```

3. After the audit policy is turned on and the database starts generating traffic:
 - a. On the Azure portal, select your Event Hub and go to the "Messages" dashboard. Select a timeframe period and check the count under the "Outgoing messages" tag. If the count is greater than 0 in the

selected timeframe, this indicates that messages are being received by the Event Hub.

- b. To check that the policy is enabled on your server, run the following command, filling in the values for <resource-group> and <sql-server-name>:

```
az sql server audit-policy show --resource-group <resource-group> --name <sql-server-name>
```

For more information, see [az sql server audit-policy](#).

- c. On the Agentless Gateway host(s), check the \$JSONAR_DATADIR/eventhub/<eventhub_name> folder to see that *.json files are being received.

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.
- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
- [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
- [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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